

© Brendan Choi 2021

B. Choi, *Introduction to Python Network Automation*

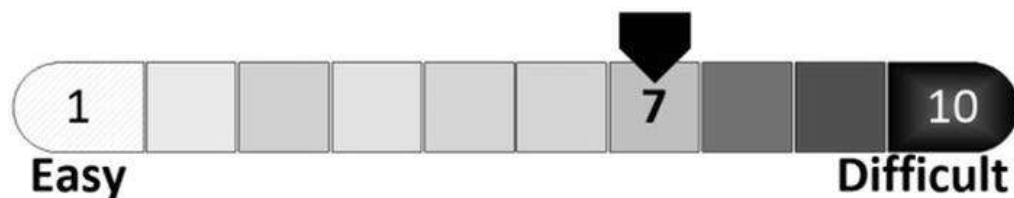
[https://doi-org.ezproxy.sfpl.org/10.1007/978-1-4842-6806-3\\_12](https://doi-org.ezproxy.sfpl.org/10.1007/978-1-4842-6806-3_12)

## 12. Building a Python Automation Lab Environment

Brendan Choi<sup>1</sup>

(1) Sydney, NSW, Australia

We have been test-driving an old Cisco device running an outdated IOS version (12.x) in our Telnet labs so far. To test newer features on newer IOS versions and equip our lab with the switching (L2) functionalities, we need to integrate Cisco CML-PE's L2 and L3 images into the GNS3 lab environment. This chapter will guide you through Cisco CML-PE L2 and L3 image integrations on GNS3. You will be guided through each step to download and integrate the L2 and L3 images in your GNS3 environment so you can conduct test with newer IOS feature sets in your network labs in the next chapter. You will learn about CML-PE as a product, download the required images for your lab use, integrate the downloaded images on GNS3, and then build a CML-PE basic lab topology to test various Python networking labs in the following chapters.



You have come a long way in this book on your first network automation journey. Thus far, I hope you have not lost your passion for learning Python network automation. To get the most from the second half of this book, this book assumes you have thoroughly read the previous 11 chapters. In your first half of Python network automation journey, you picked up various fundamental hands-on skills required to install, run, and con-

figure an integrated virtual lab using a number of open source and proprietary software. You also started doing basic Python scripting using what you have learned so far. Of course, we could have covered many more exciting topics such as DevOps concepts, Ansible, ACI, REST API, HTML, Postman, YANG, NETCONF, and much more. However, this book's purpose is clear: we have been focusing on the quintessential tasks of getting you started with Python programming, which is the foundational skill of all Python application development engineers. You have probably heard the expression “Give a man a fish, and you feed him for a day; teach a man to fish, and you feed him for a lifetime.” Borrowing this great idiom for Python network application development, “Teach a network engineer how to use Ansible, and you make him a driver; teach him how to write code in Python, and you turn him into a mechanic for a lifetime.” Hence, it is your responsibility to expand on the skills you acquire from reading this book; only you can decide where you want to go next and how to apply the new skills to your work or study.

So far, you have learned various IT skills to prepare you for the following chapters that predominantly focus on Python network automation labs. If you followed along nicely without any technical hiccups up to now, you should be able to go through the remaining chapters with ease. If you have not fully set up your lab environment or if you only partially completed exercises in each chapter, I encourage you to go back and finish, as you will need all of those basics here and throughout the rest of the book.

In this chapter, we will start from where we left off at the end of Chapter **11**, completing the basic cmlab-basic lab topology built to start writing Python code and testing the code for network automation application development. Suppose you are an experienced network engineer with many years of experience on Cisco, Juniper, Arista, and HP network devices. In that case, you will have no issue following the contents of this lab. However, even if you are a new network engineer just starting your career, you will still be able to follow along with all the lab steps, as the labs are written with minimal networking concepts and come with enough explanations for training purposes. If you concentrate, you will be able to complete all the labs with little difficulty.

## Cisco CML-PERSONAL Software License Information and Software Downloads

Cisco sells Cisco Modeling Labs – Personal (CML-PERSONAL) with a yearly subscription. If you are a network engineer or a networking student who wants to explore everything that Cisco CML-PERSONAL has to offer, purchasing the subscription is the way to go. In the first year of subscription, you are required to pay US \$199, and you will continue to pay the same yearly subscription fee designated by the Cisco Learning Team. For a professional engineer who works on Cisco technologies for a living, the subscription fee is cheap compared to how much traditional network engineers had to spend on their old 3700 and 2600 series routers and 2950 series switches to break into this industry. Anyway, if you can get your hands on CML images, there are more cost-effective ways of setting up your lab, like integrating them into GNS3, which is the topic of this chapter.

As mentioned in the first chapter, CML-PERSONAL (renamed VIRL-PE, for Virtual Internet Routing Lab Personal Edition) is a virtual equipment emulator program for studying Cisco's technologies, so you can say it is vendor-specific. However, the networking concepts you will learn from this Cisco tool will be universal and replicable to other vendor technologies. Once you are subscribed to Cisco CML-PERSONAL, you can download and install Cisco's CML-PERSONAL software and use routers, switches, and firewall CML-PERSONAL images to emulate the various Cisco labs. With no surprise, this book does not use CML-PERSONAL PE but uses a hybrid lab setup using Cisco CML-PERSONAL's router (L3) and switch (L2) images on GNS3. The CML-PERSONAL images installation and integration procedures on GNS3 are almost identical to the Cisco router IOS image integration on GNS3. Once again, your Windows host PC needs a connection to the Internet. CML-PERSONAL is Cisco's proprietary software, and free distribution of CML-PERSONAL software and CML-PERSONAL images are prohibited by law. If you can afford the subscription, we recommend you subscribe to CML-PERSONAL at

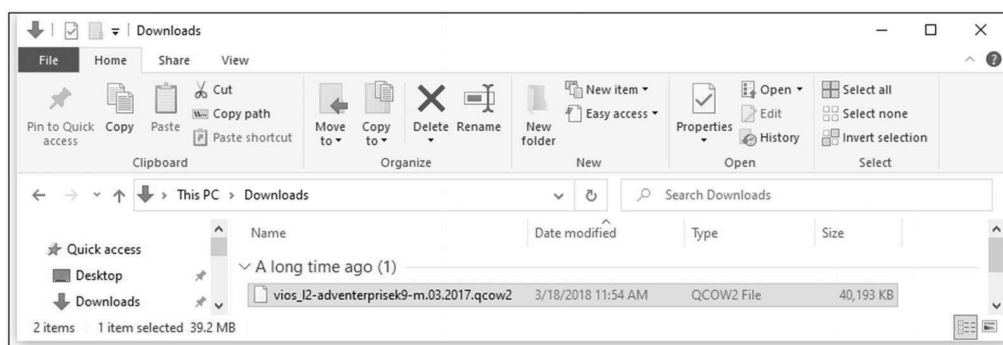
<https://learningnetworkstore.cisco.com/cisco-modeling-labs-personal/cisco-cml-personal>.

This book makes a disclaimer that any Cisco IOS, CML-PERSONAL, or any other proprietary software purchase and use are your responsibilities. No software is provided to you as part of this book. There are several ways to locate the software mentioned in this book. Please refer to <https://developer.cisco.com/modeling-labs/> for more information.

## Downloading the Cisco CML-PERSONAL IOSvL2 (Switch) Image

Before we can integrate the Cisco CML-PERSONAL L2 switch image on GNS3, you will need to locate, download, and save it to your host PC's Downloads folder. You can get information on the CML-PERSONAL L2 switch from the official GNS3 website, which will take you to Cisco's CML-PERSONAL L2 site (<https://learningnetwork.cisco.com/s/article/iosvl2-more-info-updated-10-2-15-x>).

The name of the CML-PERSONAL switch image used in this book is `vios_l2-adventerprisek9-m.03.2017.qcow2`. You can use the same one or any newer images you can find and download. After downloading the switch image file, save it to the Downloads folder. Saving it to the Downloads folder is important to save time as GNS3 will first look for the image in the Downloads folder (see Figure 12-1).



**Figure 12-1.** CML-PERSONAL L2 switch image, saved to the Downloads folder

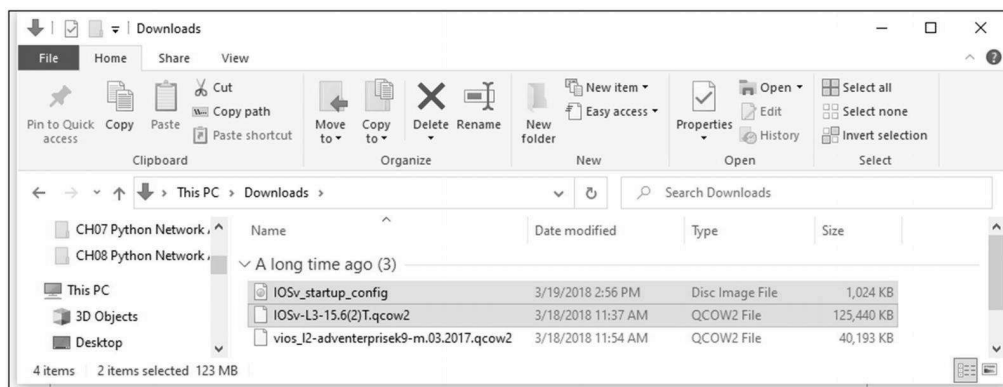
## Downloading the Cisco CML-PERSONAL IOSv (Router) Image and startup\_config Files

To integrate a CML-PERSONAL L3 or router image, you have to download the working IOSv L3 Qemu (`qcow2`) and `IOSv_startup_config.img` image files.

Unlike the L2 switch, the CML-PERSONAL router image integration with GNS3 also requires the `IOSv_startup_config.img` file for device startup, which can be downloaded from CML-PERSONAL's official site or `sourceforge.net`. Information on the GNS3 CML-PERSONAL L3 router and `IOSv_startup_config.img` can be obtained from Cisco's CML-PERSONAL site. Go to the following suggested sites to find out more about the Cisco CML-PERSONAL L3 software:

- *GNS3 CML-PERSONAL IOSv:*  
<https://learningnetwork.cisco.com/s/article/iosv-more-info-updated-4-20-15-x>
- *CML-PERSONAL IOSv\_startup\_config file download:*  
<https://sourceforge.net/projects/gns-3/files/Qemu%20Appliances/>

The CML-PERSONAL IOSv or L3 router image name used in this book is `IOSv-L3-15.6(2)T.qcow2`; this is an old image, but it still meets our needs. Since we are not exploring all the networking features, Cisco devices only serve our needs to learn the network automation concepts and Python application developments. We can use any version of CML L2 and L3 images as long as it can carry the traffic and saves us time and resources from running full-blown hardware-based labs. Although we are focusing on network automation, the routing and switching concepts are not the core of this book. So, Cisco CML-PERSONAL images provide us with the basic L2 and L3 connections to carry traffic and learn Python scripting skills using Telnet, SSH, RESTCONF, and other Python libraries. Download the two required files, as shown in Figure 12-2, and save them in the `Downloads` folder.



**Figure 12-2.** CML-PERSONAL L3 router image, saved to the Downloads folder

Once you have the vIOS L2 image, vIOS L3 image, and vIOS startup-config files, you are ready to install and integrate them to your GNS3.

## Installing the Cisco CML-PERSONAL L2 Switch and CML-PERSONAL L3 on GNS3

At the end of Chapter [11](#), we copied and created a new project called `cml-lab-basic`. Let's reopen that project as this is the starting point of our tasks in this chapter. If you are continuing your work from Chapter [11](#), continue working from GNS3, but if you have just powered on your PC, then first launch GNS3 using the GNS3 shortcut icon on your desktop. If the lab environment was set up correctly, GNS3 runs, and the GNS3 VM installed on VMware Workstation should start automatically after a few seconds.

 **Installation tip** If the CML-PERSONAL installation does not work as described due to a software failure, you must delete the installed switch or router templates by right-clicking GNS3 and trying to install the images again. CML-PERSONAL images are not made for GNS3, so sometimes they will not integrate to GNS3 on the first try. If the CML-PERSONAL image does not get installed properly after the first attempt, do not give up; keep trying until it is installed properly.

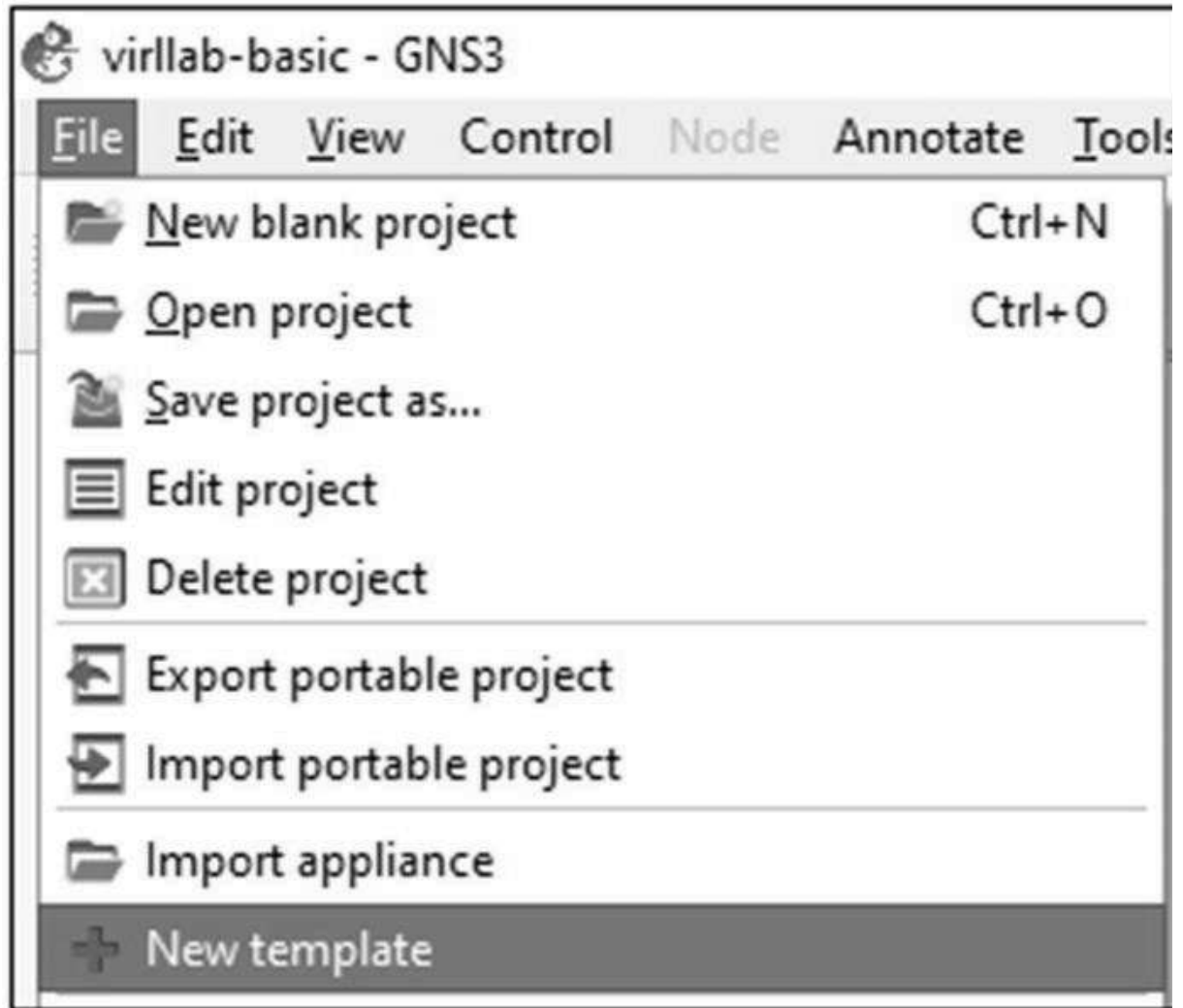
## Installing the Cisco CML-PERSONAL L2 Switch on GNS3

You will install the Cisco CML-PERSONAL L2 switch image on GNS3 first. For correct installation, the host PC must be connected to the Internet. After this installation, you will run the switching labs using the installed image, which was not possible on the older IOS image integration.

| # | Task |
|---|------|
|---|------|

|   |                                                                                                  |
|---|--------------------------------------------------------------------------------------------------|
| 1 | In the GNS3 main window, go to File and click + New template . See Figure <a href="#">12-3</a> . |
|---|--------------------------------------------------------------------------------------------------|

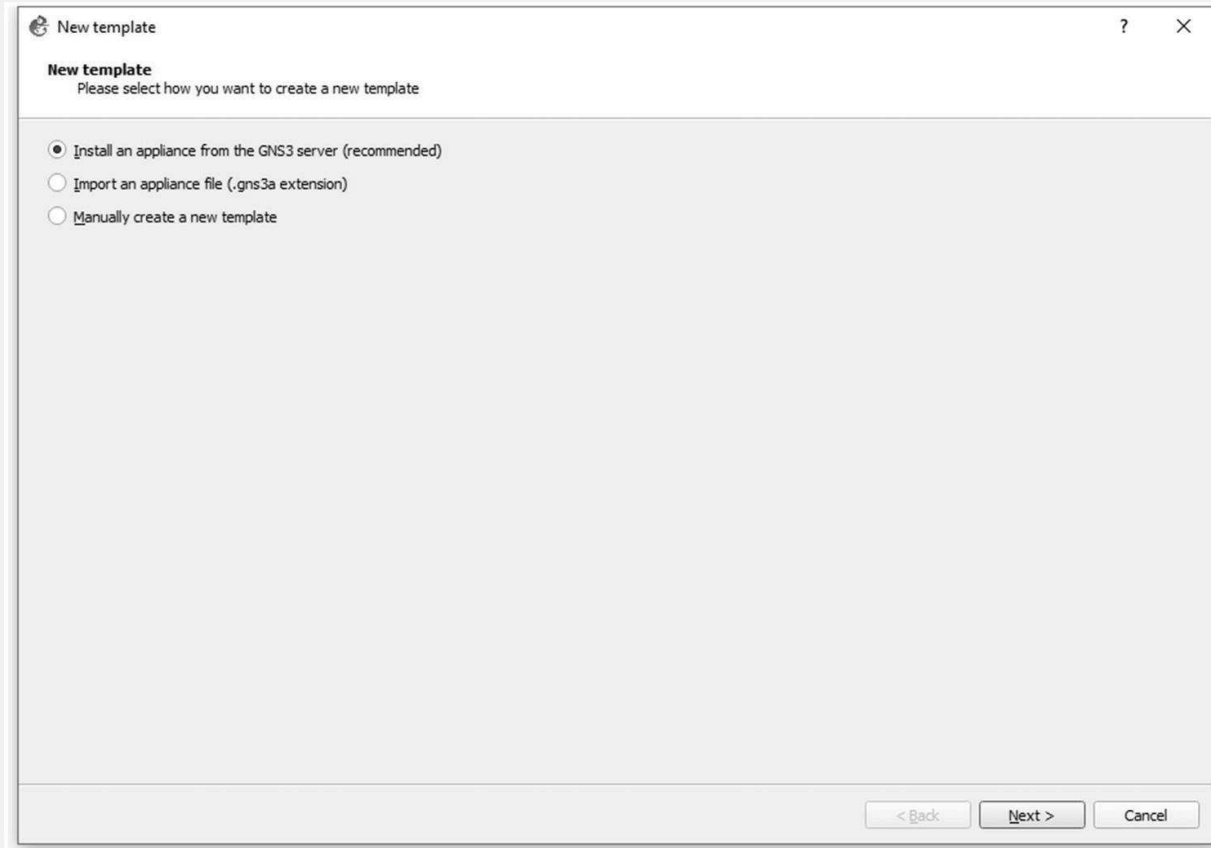
## # Task



**Figure 12-3.** CML L2 switch installation, adding a new template

- 2 When the “New template” wizard opens , leave the default selection of “Install an appliance from the GNS3 server (recommended)” and click Next. See Figure [12-4](#).

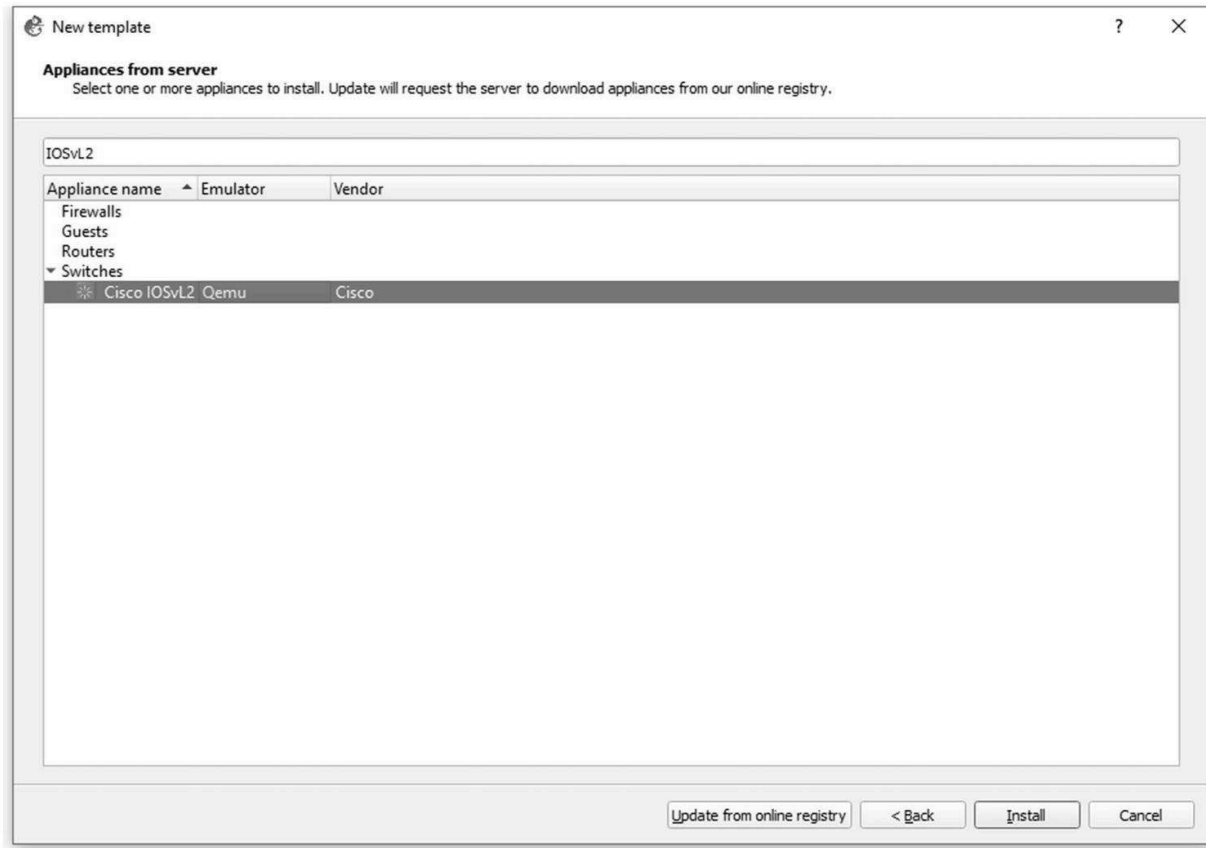
## # Task



**Figure 12-4.** CML L2 switch installation, new template selection

- 3 In the “Appliances from server” window , in the search field, type in **IOSvL2** to locate Cisco IOSvL Qemu under Switches. Select the searched device and click Install; this is the template for the Cisco CML-PERSONAL L2 switch on GNS3. See Figure **12-5**.

## # Task

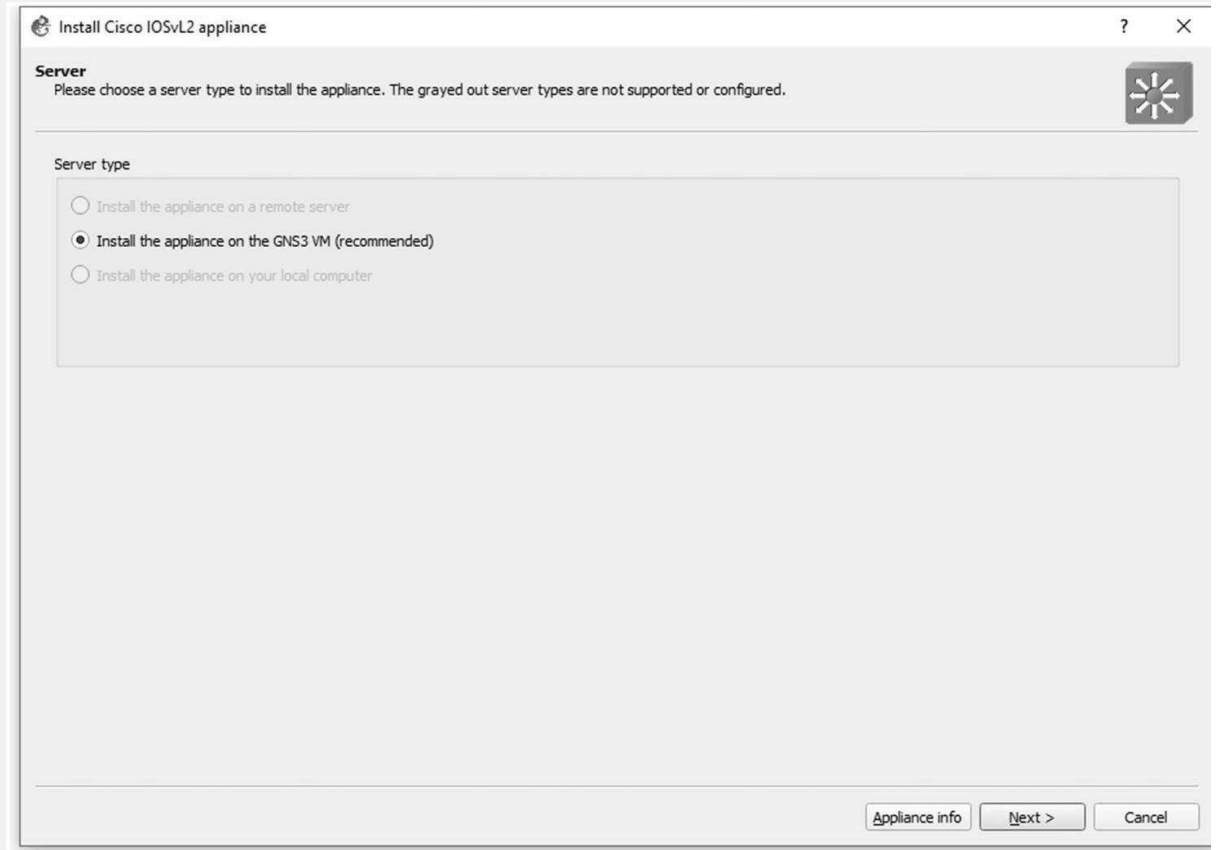


**Figure 12-5.** CML L2 Switch installation, appliances from server

4

In the Server window , click Next. See Figure [12-6](#).

## # Task

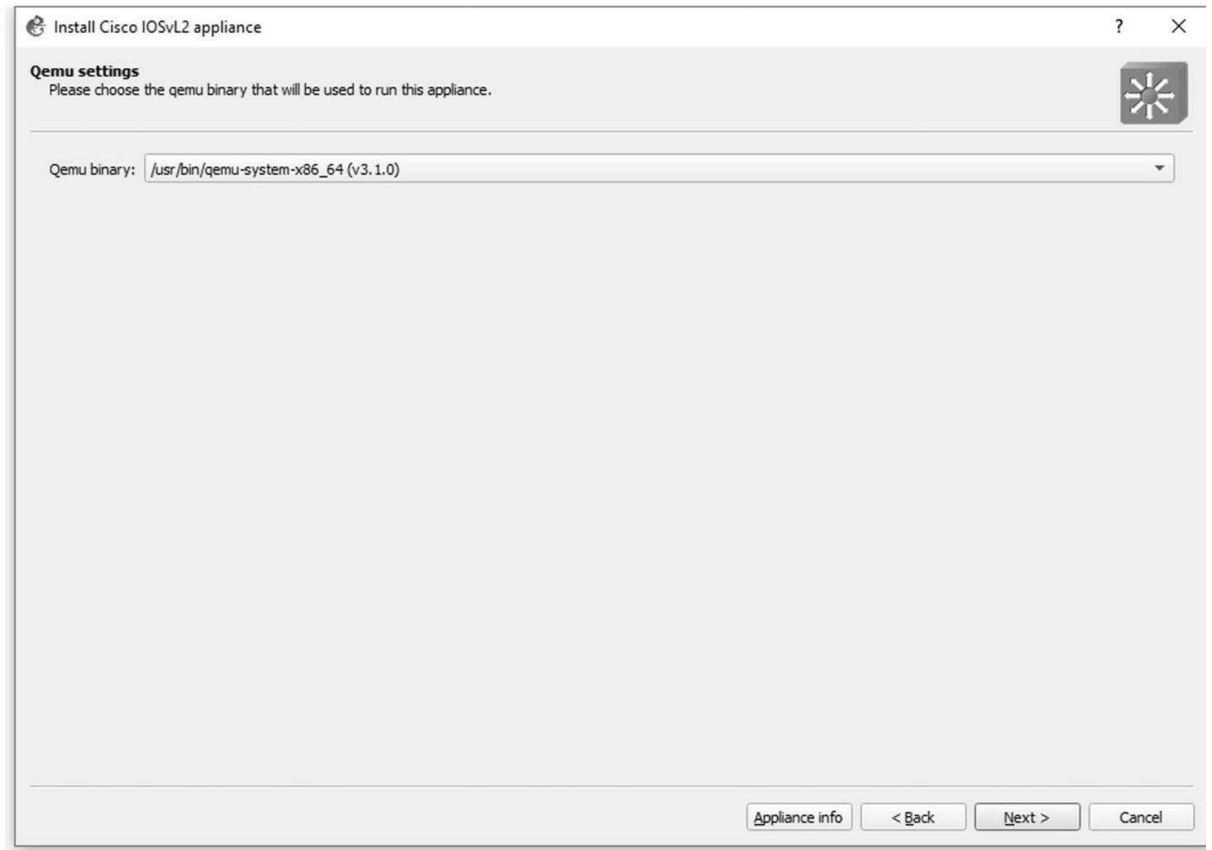


**Figure 12-6.** CML L2 switch installation, server

5

In the “Qemu settings” window , click Next. See Figure [12-7](#).

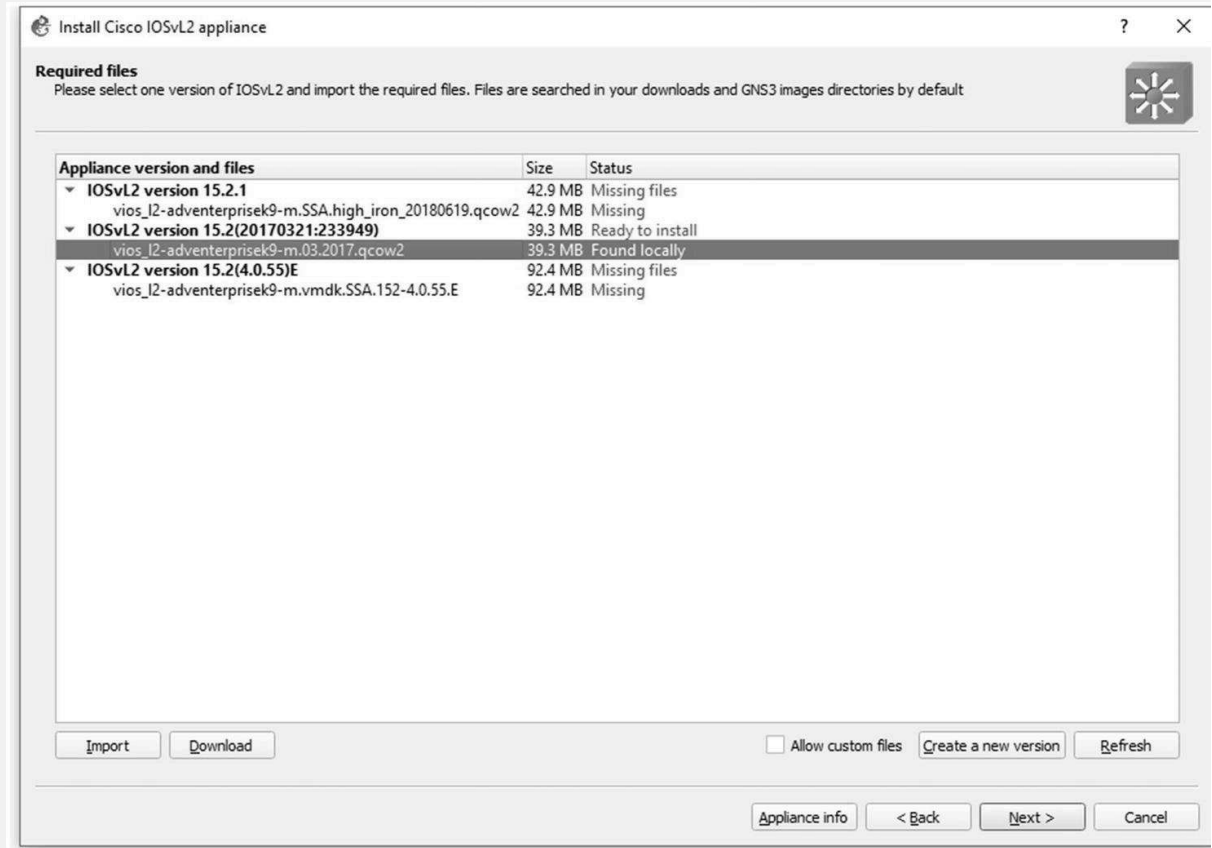
## # Task



**Figure 12-7.** CML L2 switch installation, Qemu settings

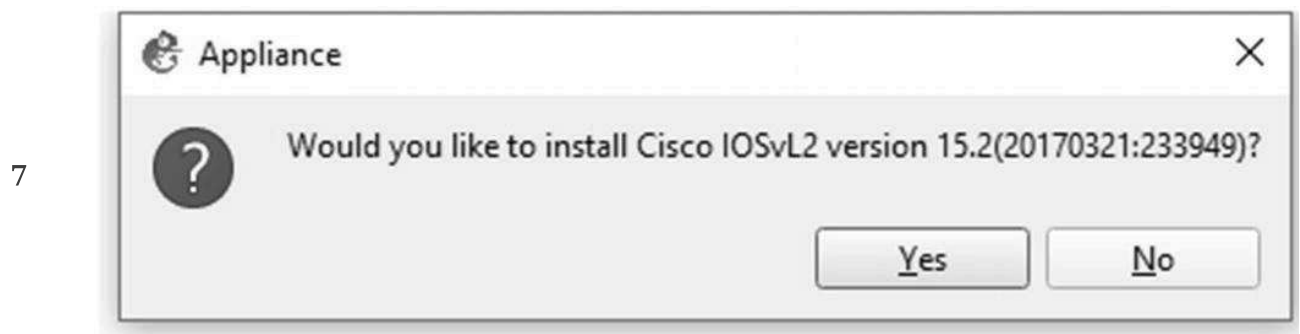
- 6 Under “Required files,” if you have placed the CML-PERSONAL L2 .qcow2 file in the Downloads folder, the installation wizard will automatically detect the L2 image automatically and display it in green. You can also see the missing files in red. Highlight your image and click Next. See Figure 12-8.

## # Task



**Figure 12-8.** CML L2 switch installation, required files

When the Appliance window pops up , click Yes. See Figure [12-9](#).

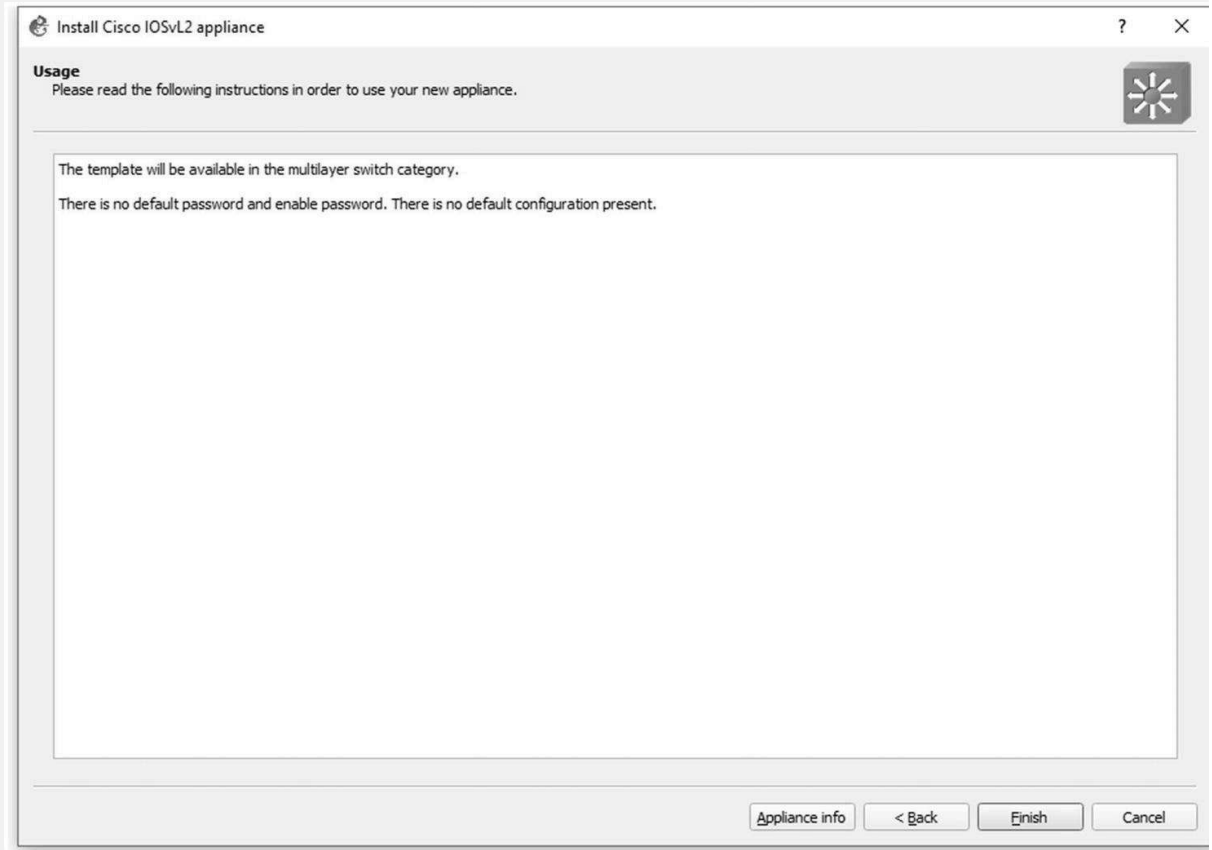


**Figure 12-9.** CML L2 installation, appliance install pop-up

8

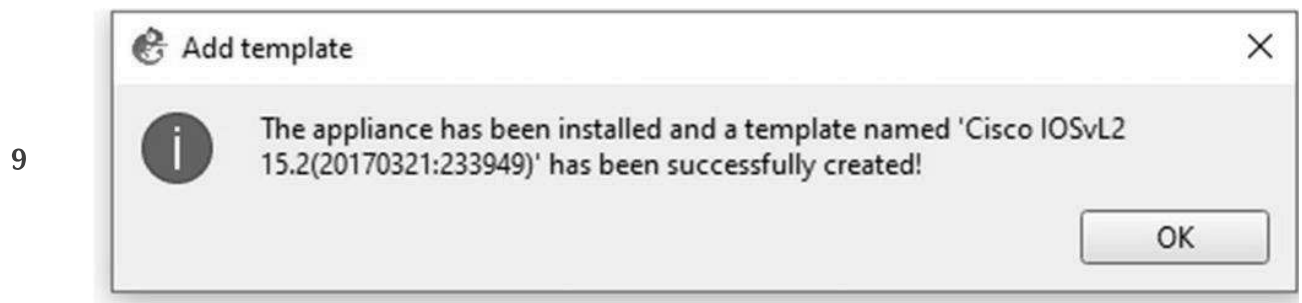
To complete the installation , click Finish. See Figure [12-10](#).

## # Task



**Figure 12-10.** CM L2 switch installation, usage

The “Add template” window pops up ; click OK to finish the installation. See Figure [12-11](#).



**Figure 12-11.** CML L2 switch installation, “Add template” pop-up

- 10 On GNS3, click the Switch button on the left, which looks like two opposite arrows, to open the switches. See Figure [12-12](#).

# Task



**Figure 12-12.** CML L2 switch installation, select switch icon on GNS3

## # Task

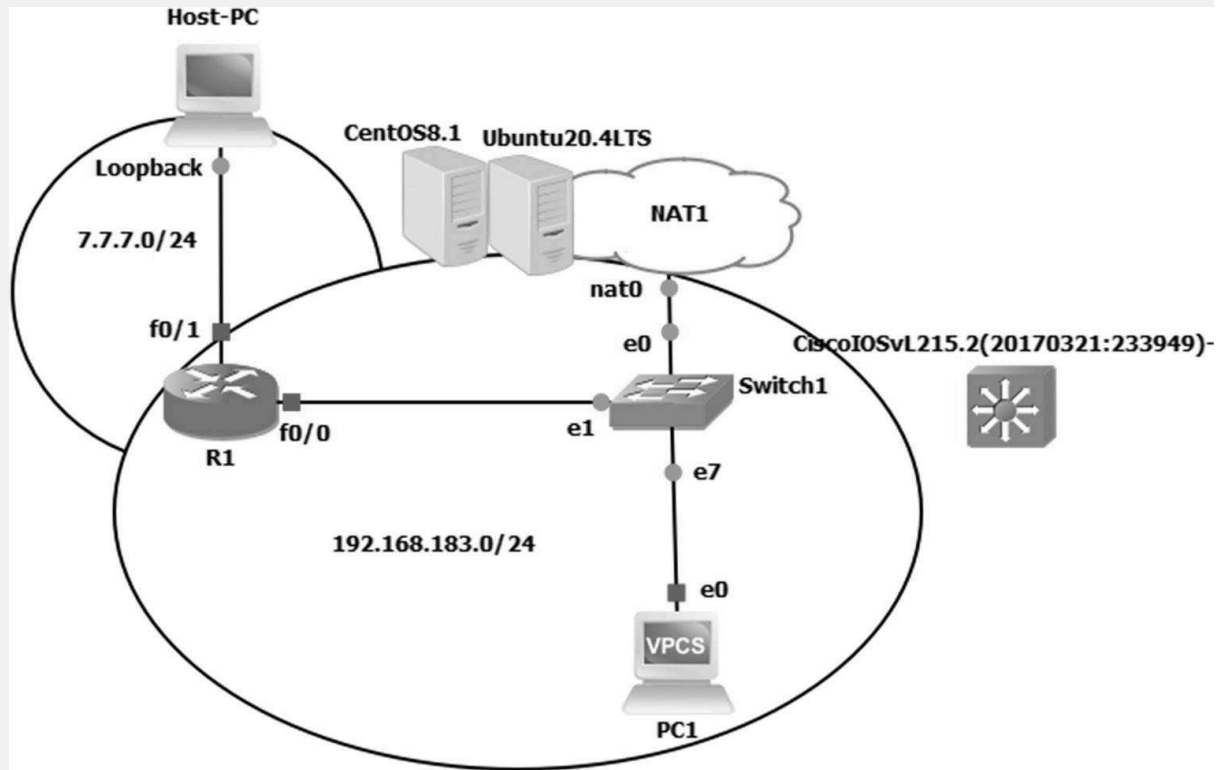
If Cisco CML-PERSONAL IOSvL2 was installed correctly, you would see your new switch devices installed under Switches. See Figure [12-13](#).



**Figure 12-13.** CML L2 switch installation, selecting the Cisco IOSvL2 switch

Now, drag and drop the IOSvL2 switch to the cmlab-basic lab, as shown in Figure [12-14](#).

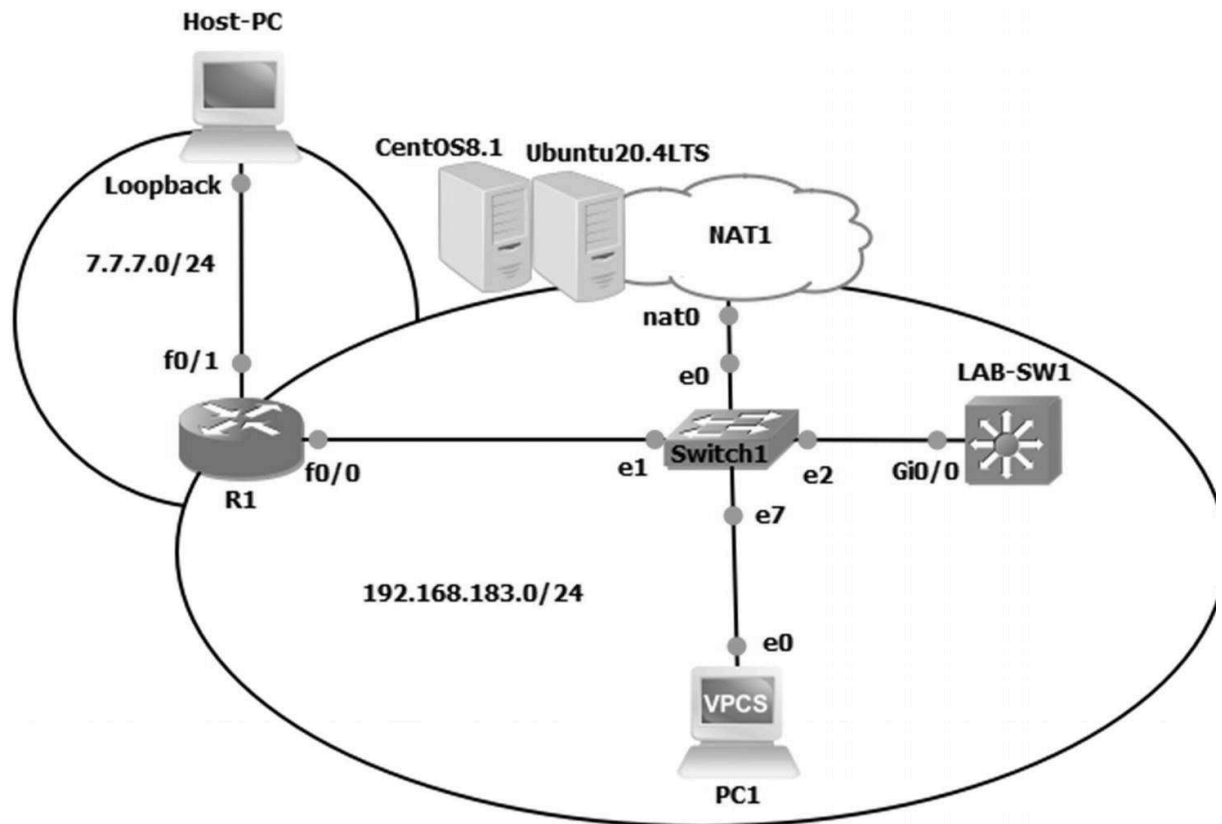
## # Task



**Figure 12-14.** CML L2 switch installation, dragging and dropping a CML switch

- 13 Using the Add Link tool, connect Gi0/0 of the new switch to Switch1's e2 interface, as shown in Figure [12-15](#). Then power on all devices on the topology and also the CentOS8.1 and Ubuntu 20.04 LTS servers on VMware Workstation 15 Pro. Rename the switch to LAB-SW1 by right-clicking and clicking the Change Hostname menu.

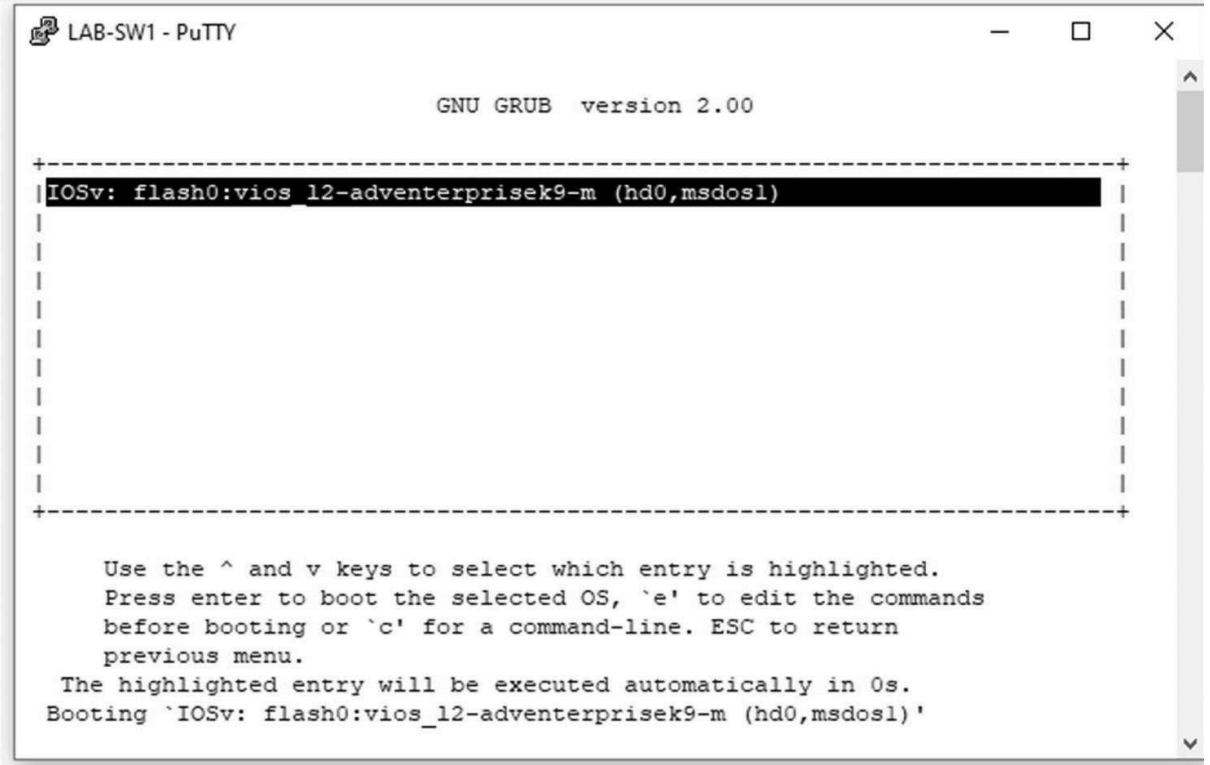
## # Task



**Figure 12-15.** CML L2 switch installation, connecting the CML switch to the network

- 14 After powering on LAB-SW1, open its console, and you will see a similar power-on screen as shown in Figure [12-16](#).

## # Task



**Figure 12-16.** CML L2 switch installation, powered-on CML switch screen

- 15 Your new switch's console screen should look like the following if it successfully booted up. Now you have completed the CML-PERSONAL L2 switch installation on your GNS3. Get ready for some switching labs.

```
Cisco IOS Software, vios_l2 Software (vios_l2-ADVENTERPRISEK9-M),
Experimental Version 15.2(20170321:233949) [mmen 101]
```

```
Copyright (c) 1986-2017 by Cisco Systems, Inc.
```

```
Compiled Wed 22-Mar-17 08:38 by mmen
```

```
*****
```

```
* IOSv is strictly limited to use for evaluation, demonstration and IOS
```

```
* education. IOSv is provided as-is and is not supported by Cisco's
```

**# Task**

\* Technical Advisory Center. Any use or disclosure, in whole or in part,

\* of the IOSv Software or Documentation to any third party for any

\*

\*Jan 9 10:36:48.512: %PLATFORM-5-SIGNATURE\_VERIFIED: Image  
'flash0:/vios\_l2-adventerprisek9-m' passed code signing verification pur-  
poses is expressly prohibited except as otherwise authorized by \*

\* Cisco in writing.

\*\*\*\*\*

Switch>

## Quick Communication Test on CML L2 Switch Integration on GNS3

At the end of the CML-PERSONAL L2 switch installation, quickly perform communication verification from the switch to other devices on the network. The recommended ICMP tests are described as follows:

**# Task**

- 1 Quickly log in and configure the correct switch name and a VLAN 1 interface on the new switch with a valid IP address.

Switch>**enable** # Enter Privileged EXEC mode

Switch#**configure terminal** # Enter Global Configuration mode

LAB-SW1(config)#**hostname LAB-SW1** # Rename hostname

## # Task

```
LAB-SW1(config)#spanning-tree vlan 1 root primary #
Designate this switch as the root primary
```

```
LAB-SW1(config)#interface vlan 1 # Enter interface vlan
1 config mode
```

```
LAB-SW1(config-if)#ip add 192.168.183.101
255.255.255.0 # Configure an IP address
```

```
LAB-SW1(config-if)#no shut # Bring up this interface
```

```
LAB-SW1(config-if)#end # Go back to Privileged EXEC
mode
```

- On LAB-SW1 , as soon as you bring up Gi0/0, you may be prompted with duplex mismatch errors from R1's f0/0 interface.
- 2 To stop this annoying CDP error message on LAB-SW1, you need to hard-code R1's f0/0 interface speed to 100 and duplex to full. If you do not get this error, you can continue to the next step.

```
LAB-SW1(config-if)#
```

```
*Aug 2 11:16:38.977: %CDP-4-DUPLEX_MISMATCH: duplex
mismatch discovered on GigabitEthernet0/0 (not half
duplex), with R1 FastEthernet0/0 (half duplex).
```

Open console for R1 and change the interface speed and duplex as follows:

```
R1#configure terminal # Enter Global Configuration mode
```

```
R1(config)#interface f0/0 # Enter interface configura-
tion mode
```

**# Task**

```
R1(config-if)#duplex full # Change interface duplex to
full (default auto)
```

```
R1(config-if)#speed 100 # Change interface speed to 100
```

```
R1(config-if)#do write memory # save the configuration
```

- 3** Go to PC1 and run the `ip dhcp` command to get an IP address, or if you are continuing your lab from the previous chapter, run the `show ip` command to check the IP address of PC1.

```
PC1> ip dhcp # Request an IP address from DHCP server
```

```
DDORA IP 192.168.183.129/24 GW 192.168.183.2
```

```
PC1> show ip # Display IP Settings of PC1 (VPCS)
```

```
NAME          : PC1[1]
```

```
IP/MASK       : 192.168.183.129/24
```

```
GATEWAY      : 192.168.183.2
```

```
DNS          : 192.168.183.2
```

```
DHCP SERVER  : 192.168.183.254
```

```
DHCP LEASE   : 1713, 1800/900/1575
```

```
DOMAIN NAME  : localdomain
```

```
MAC          : 00:50:79:66:68:00
```

```
LPORT       : 20012
```

## # Task

```
RHOST:PORT : 127.0.0.1:20013
```

```
MTU: : 1500
```

**4** From LAB-SW1 (192.168.183.101), send ICMP messages to PC1 (192.168.183.129) and R1's f0/0 (192.168.183.133).

```
LAB-SW1#ping 192.168.183.129
```

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 192.168.183.129,  
timeout is 2 seconds:
```

```
.!!!!
```

```
Success rate is 100 percent (5/5), round-trip  
min/avg/max = 2/3/5 ms
```

```
LAB-SW1#ping 192.168.183.133
```

```
Type escape sequence to abort .
```

```
Sending 5, 100-byte ICMP Echos to 192.168.183.133,  
timeout is 2 seconds:
```

```
.!!!!
```

```
Success rate is 80 percent (4/5), round-trip  
min/avg/max = 2/4/6 ms
```

## # Task

5 Make sure that both Linux servers are up and running on VMware Workstation. From the previous chapter, we know the IP address of the `centos8s1` server is 192.168.183.130, and the `ubuntu20s1` server is 192.168.183.132.

From LAB-SW1, send ICMP packets (ping) to both `centos8s1` and `ubuntu20s1`. Your server IP address may be different from the example shown here:

```
LAB-SW1#ping 192.168.183.130
```

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.183.130, timeout is 2 seconds:

!!!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 3/6/13 ms

```
LAB-SW1#ping 192.168.183.132
```

Type escape sequence to abort .

Sending 5, 100-byte ICMP Echos to 192.168.183.132, timeout is 2 seconds:

!!!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 2/3/6 ms

## # Task

- 6 After successful communication testing, copy the running configuration to the startup configuration.

```
LAB-SW1#copy running-config startup-config # Save running configuration to startup configuration
```

```
Destination filename [startup-config]?
```

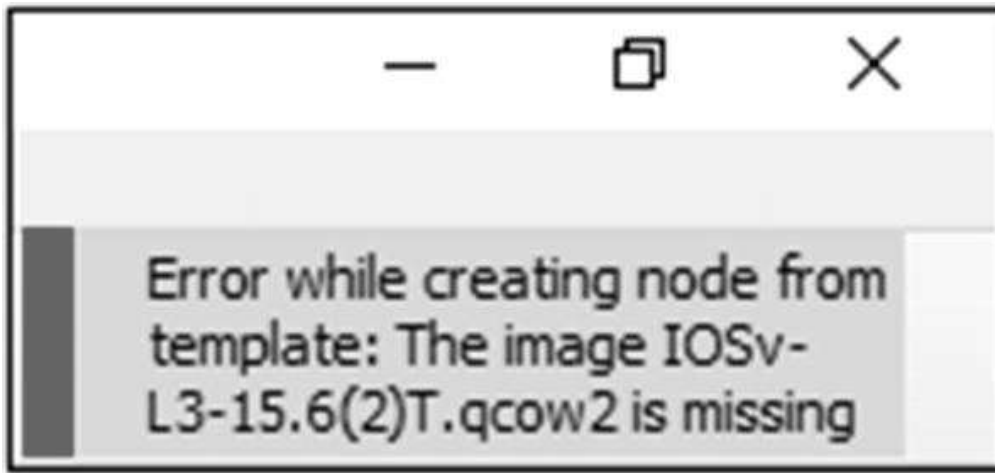
```
Building configuration...
```

```
Compressed configuration from 3559 bytes to 1604 bytes[OK]
```

You have now completed the integration and verification of the CML L2 switch image on GNS2. With this image, we can many switches to test multiple-device Python network automation labs. Next, go ahead and integrate the L3 router image.

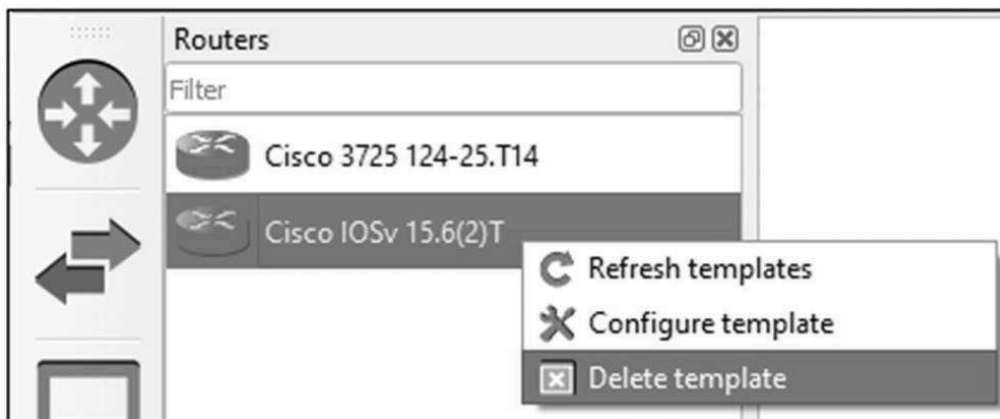
## Installing the Cisco CML-PERSONAL L3 Router on GNS3

Now you will follow the procedures shown in this section to install the CML-PERSONAL L3 router on GNS3; this will enable you to run the CML-PERSONAL router on GNS3. And as usual, you will require a connection to the Internet to complete this procedure. It is best to be connected to the home network than work during the installation process. Please note that if your host PC is connected to the internet via a company proxy, you may encounter some problems . See Figure [12-17](#).



**Figure 12-17.** GNS3, CML-PERSONAL L3 image error

⚠ The integration of CML images looks normal until you drag and drop the icon onto the GNS3 Topology canvas. If GNS3 produces a `qcow2` or other error and does not work, you may have to use the “Delete template” option as shown in Figure [12-18](#) and restart the integration process from scratch.



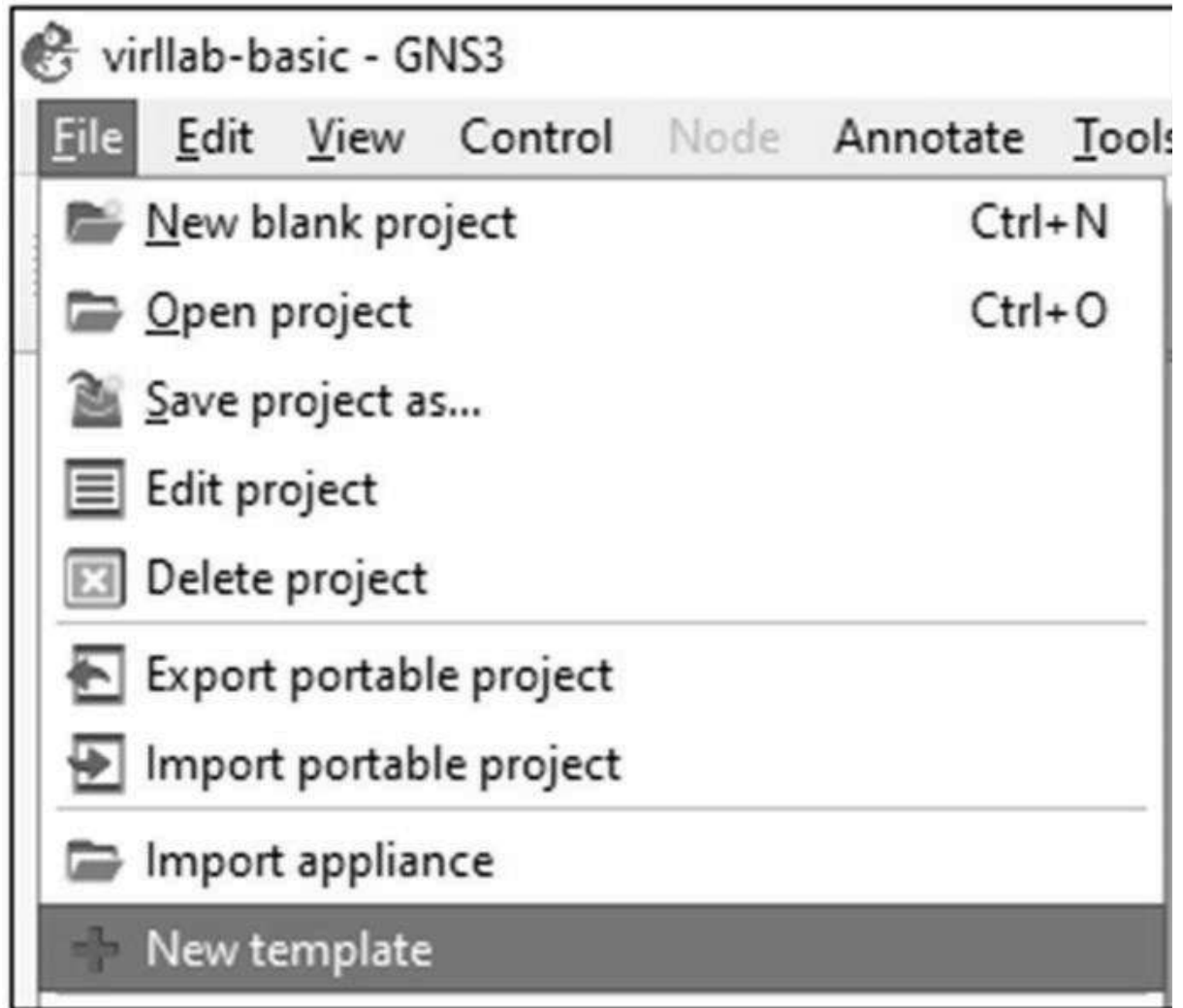
**Figure 12-18.** GNS3, deleting the CML-PERSONAL L3 template

Now add a new template for the CML L3 router image and complete the integration.

| # | Task |
|---|------|
|---|------|

- |   |                                                                                                     |
|---|-----------------------------------------------------------------------------------------------------|
| 1 | In the GNS3 main window , go to File and click “+ New template.” See Figure <a href="#">12-19</a> . |
|---|-----------------------------------------------------------------------------------------------------|

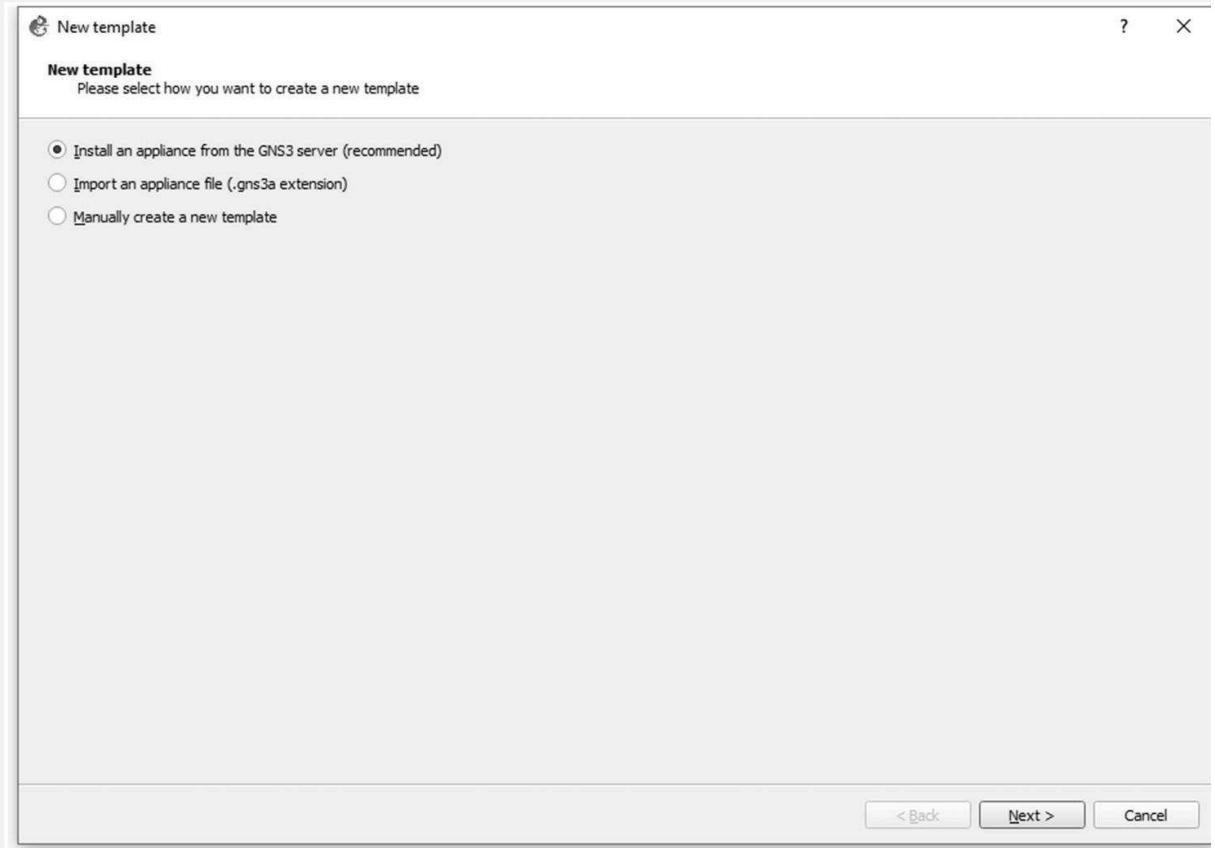
## # Task



**Figure 12-19.** CML L3 router installation, adding a new template

- 2 When the “New template ” wizard opens, leave the default selection of “Install an appliance from the GNS3 server (recommended)” and click Next. See Figure [12-20](#).

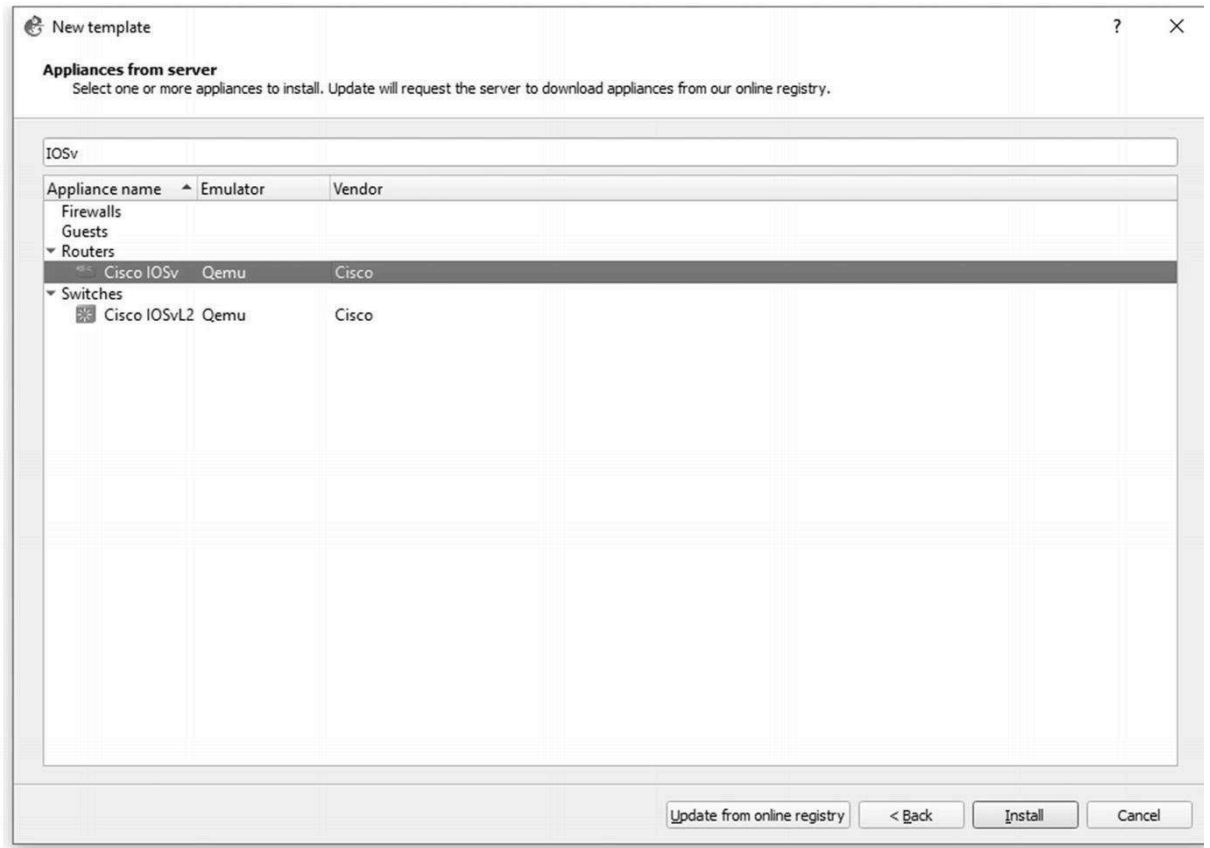
## # Task



**Figure 12-20.** CML L3 router installation, new template

- 3 In the “Appliances from server ” window, type **IOSv** in the search field to locate Cisco IOSv Qemu under Switches. Select the searched device and click Install. This is the template for the Cisco CMI PERSONAL L3 router on GNS3. See Figure [12-21](#).

## # Task

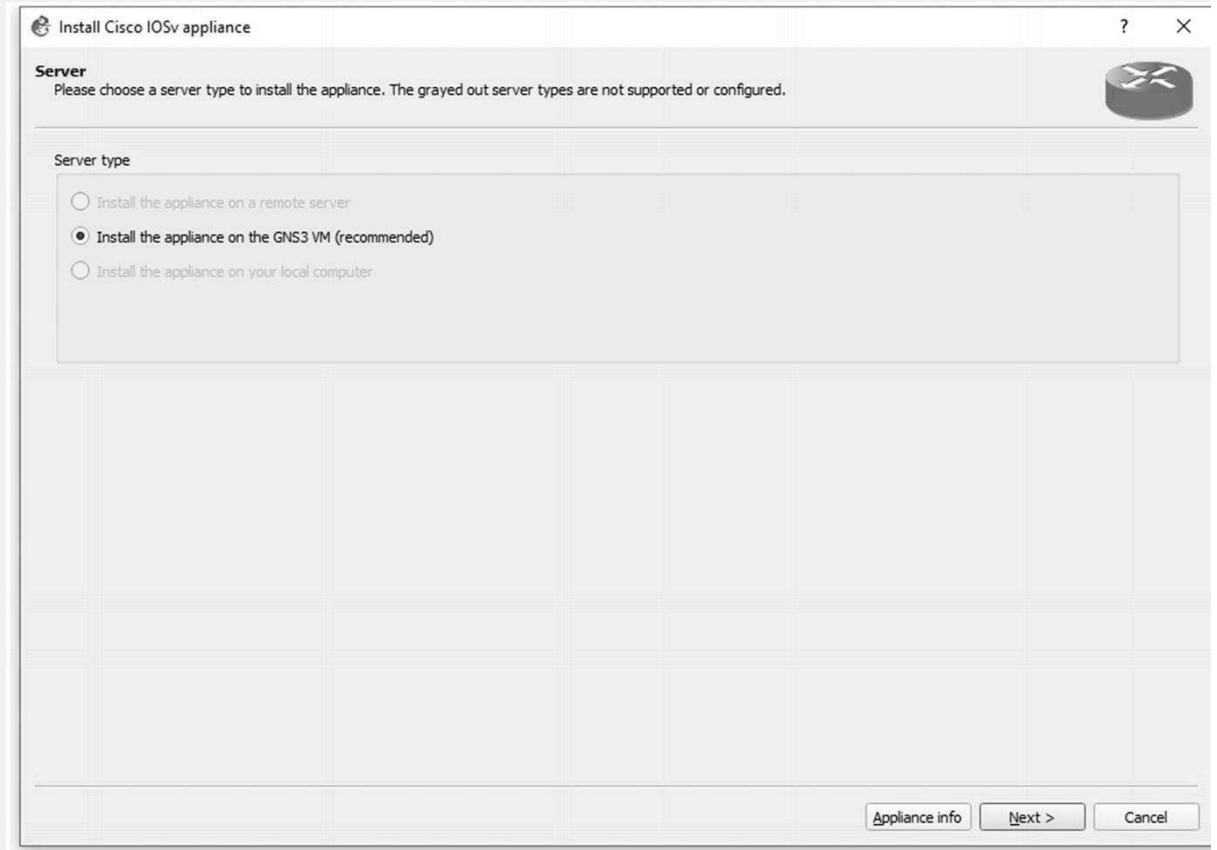


**Figure 12-21.** CML L3 router installation, appliances from server

4

In the next window , click Next. See Figure [12-22](#).

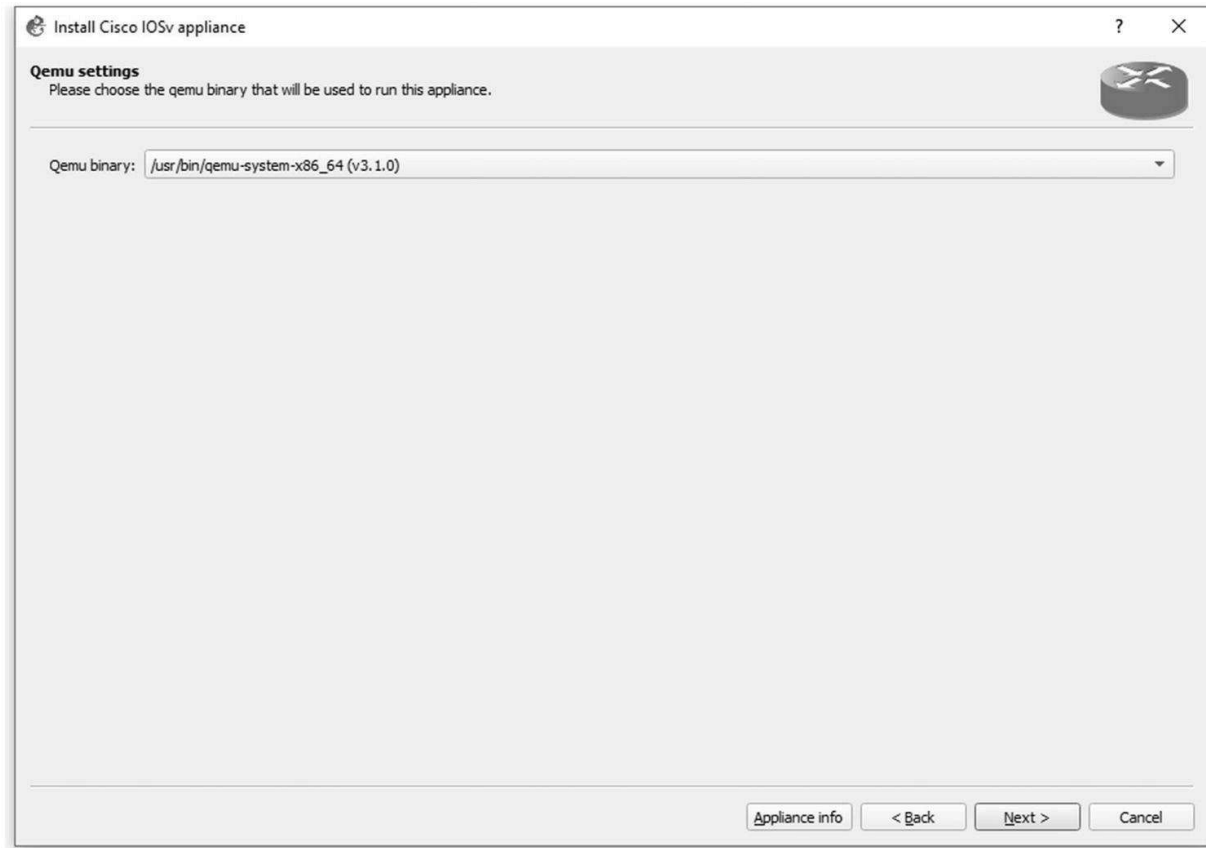
## # Task



**Figure 12-22.** CML L3 router installation, server selection

5 On the “Qemu settings” screen , click Next again. See Figure [12-23](#).

## # Task



**Figure 12-23.** CML L3 router installation, Qemu settings

- 6 It is recommended that you click the actual vIOS name under “Required files” and highlight it before clicking Next. Notice that the actual image name `vios-adventerprisek9-m.vmdk.SPA.156-2.T` is selected before clicking the Next button . See Figure [12-24](#).

# Task

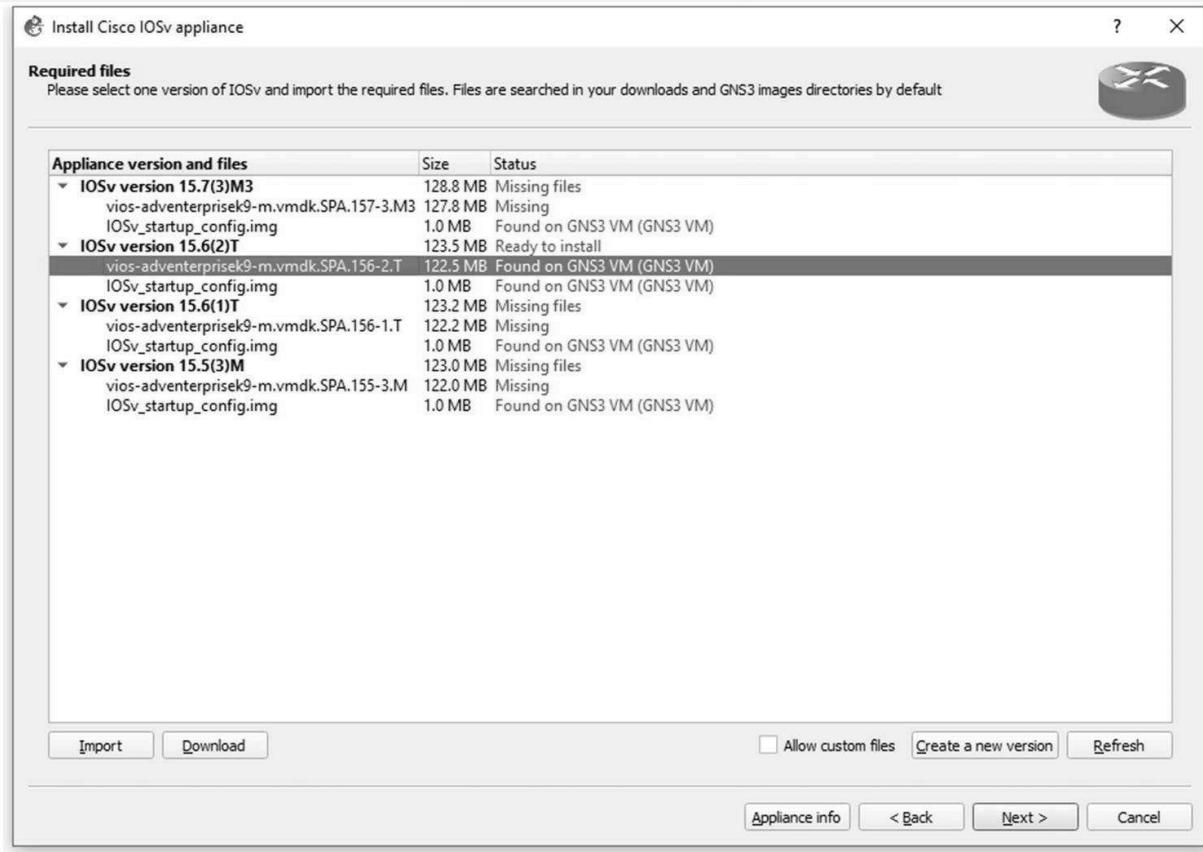


Figure 12-24. CML L3 router installation, required files

When the Appliance window pops up , click Yes. See Figure 12-25.

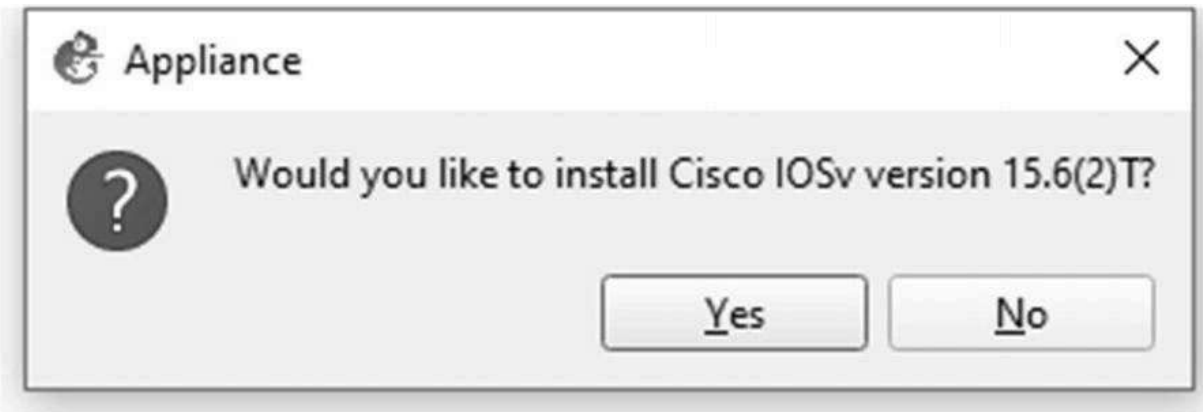
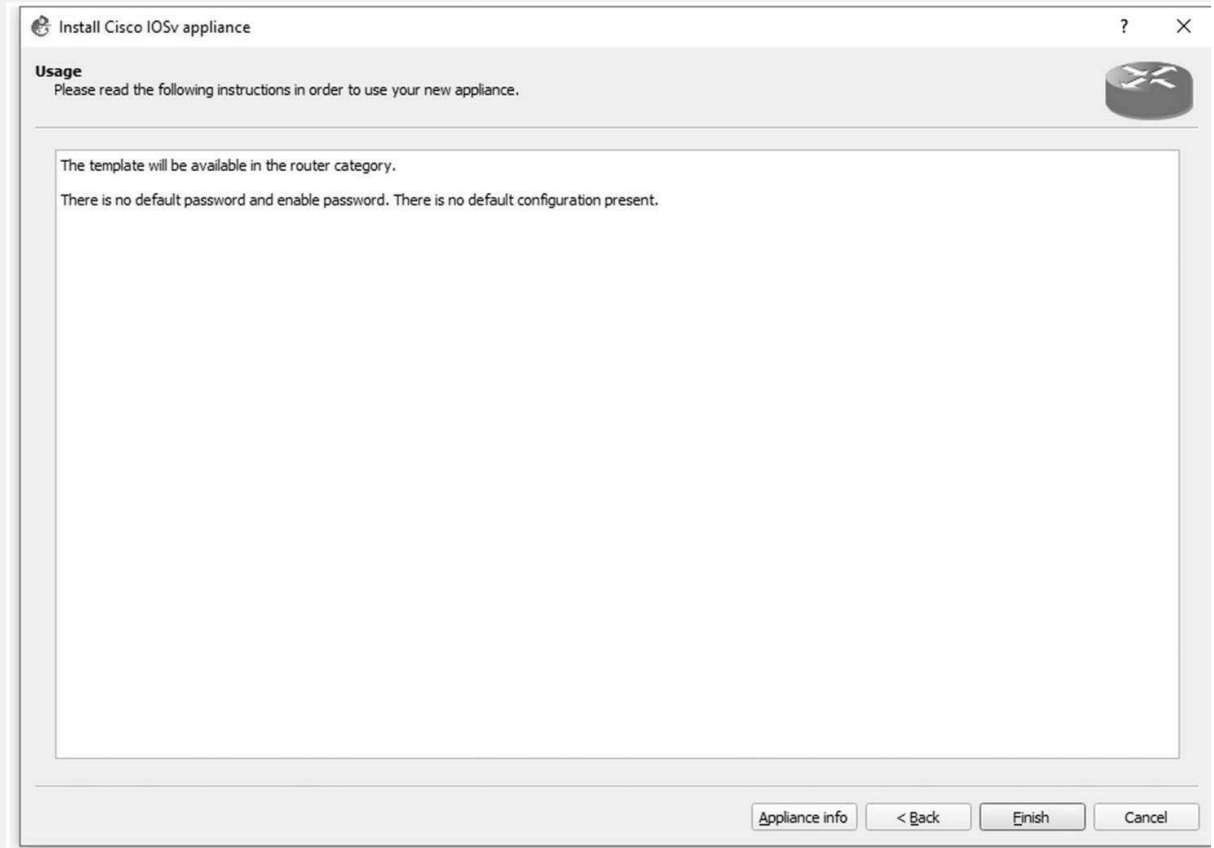


Figure 12-25. CML L3 router installation, Appliance install pop-up

8 In the Usage window , click Finish. See Figure 12-26.

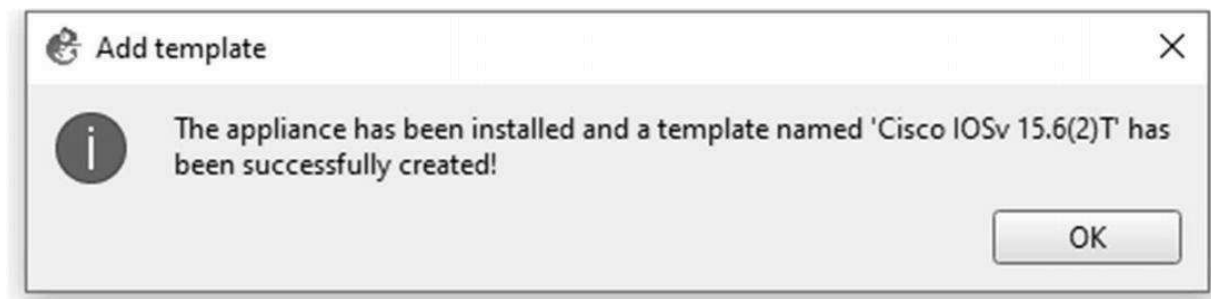
## # Task



**Figure 12-26.** CML L3 router installation, usage

When the “Add template ” message appears, click OK to complete the CML-PERSONAL L3 router image installation. See Figure [12-27](#).

9



**Figure 12-27.** CML L3 router installation, “Add template” pop-up

**10** Go back to the main user window , and this time click the Router icon that looks like a circle with four arrows. See Figure [12-28](#).

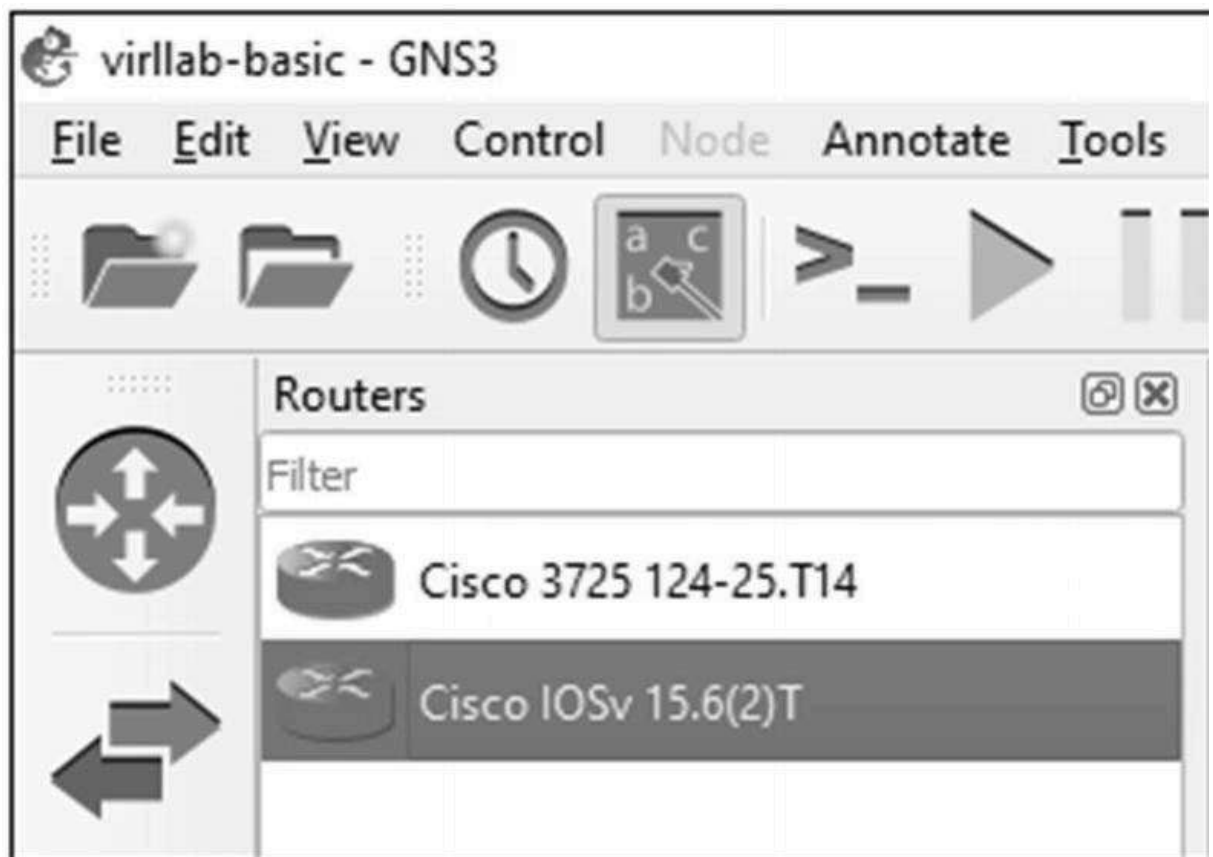
## # Task



**Figure 12-28.** CML L3 router installation, clicking the router icon on GNS3

The Routers list appears as shown in Figure 12-29, and you will immediately notice that a new IOSv router has been added.

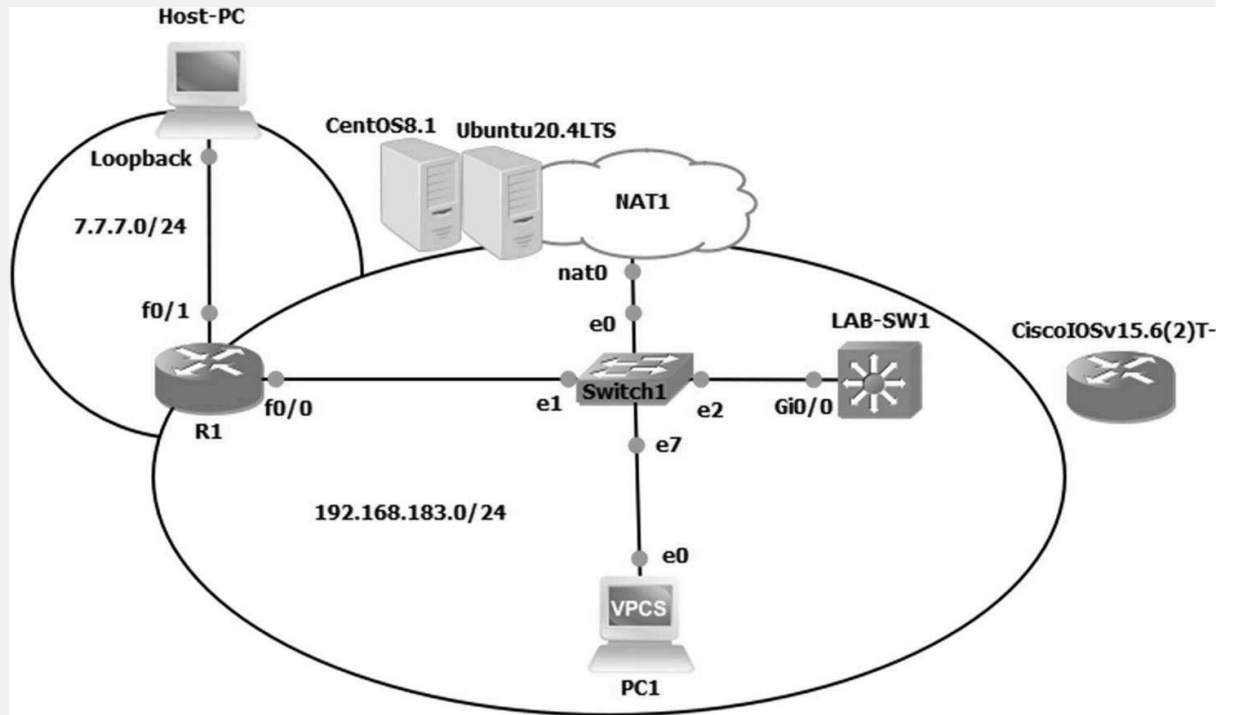
11



**Figure 12-29.** CML L3 router installation, selecting the CML router

12 Now drag and drop the new router onto the Topology canvas of the `cmlab-basic` project. Rename the router as LAB-R1. See Figure 12-30.

## # Task



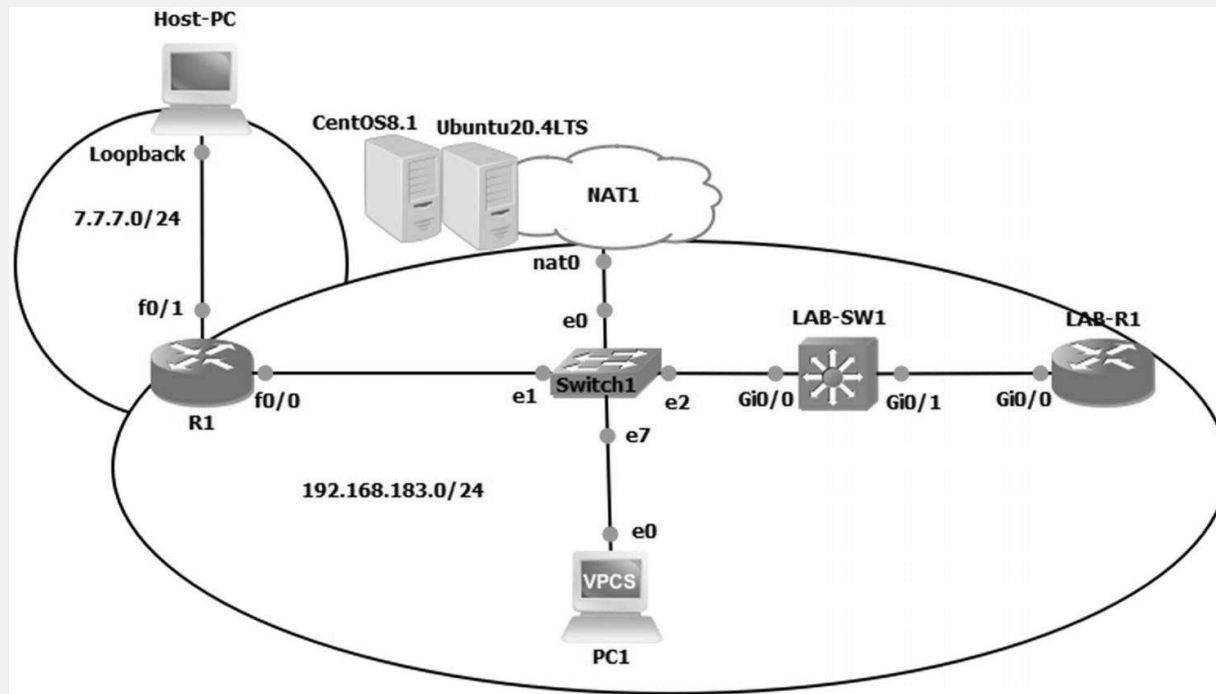
**Figure 12-30.** CML L3 router installation, dragging and dropping the CML router

13

Once the new router has been renamed, use the Add Link (connector) tool, and connect Lab-R1's Gi0/0 interface to LAB-SW1's Gi0/1 interface . You can make the connection while the other device is switched on. Now power on LAB-R1 and open the console window. See Figure [12-31](#).

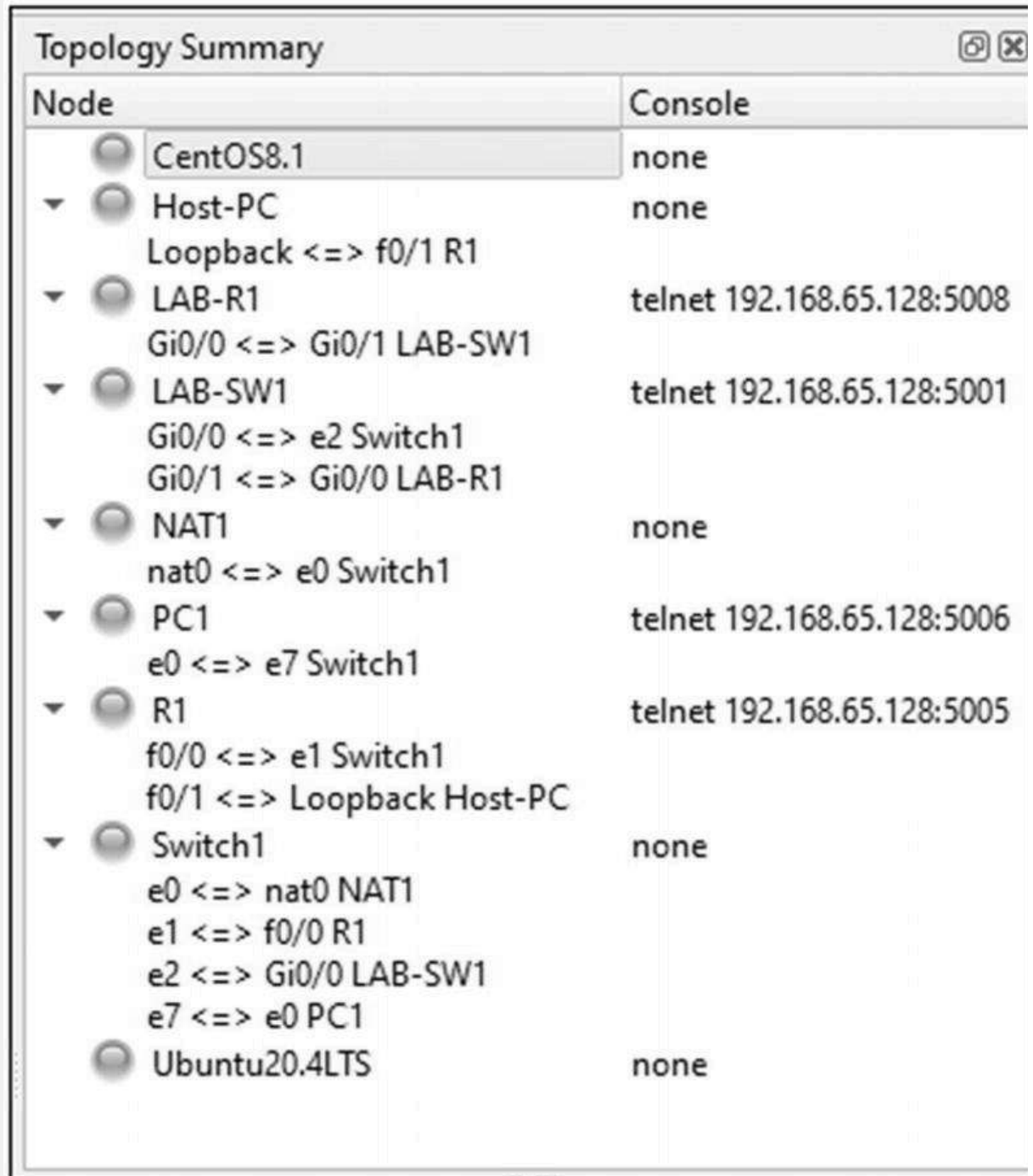
The Topology Summary window shows the virtual connection between devices (see Figure [12-32](#)).The Servers Summary window shows where each device is hosted (see Figure [12-33](#)).

## # Task



**Figure 12-31.** CML L3 router installation, connecting and starting the CML router

# Task

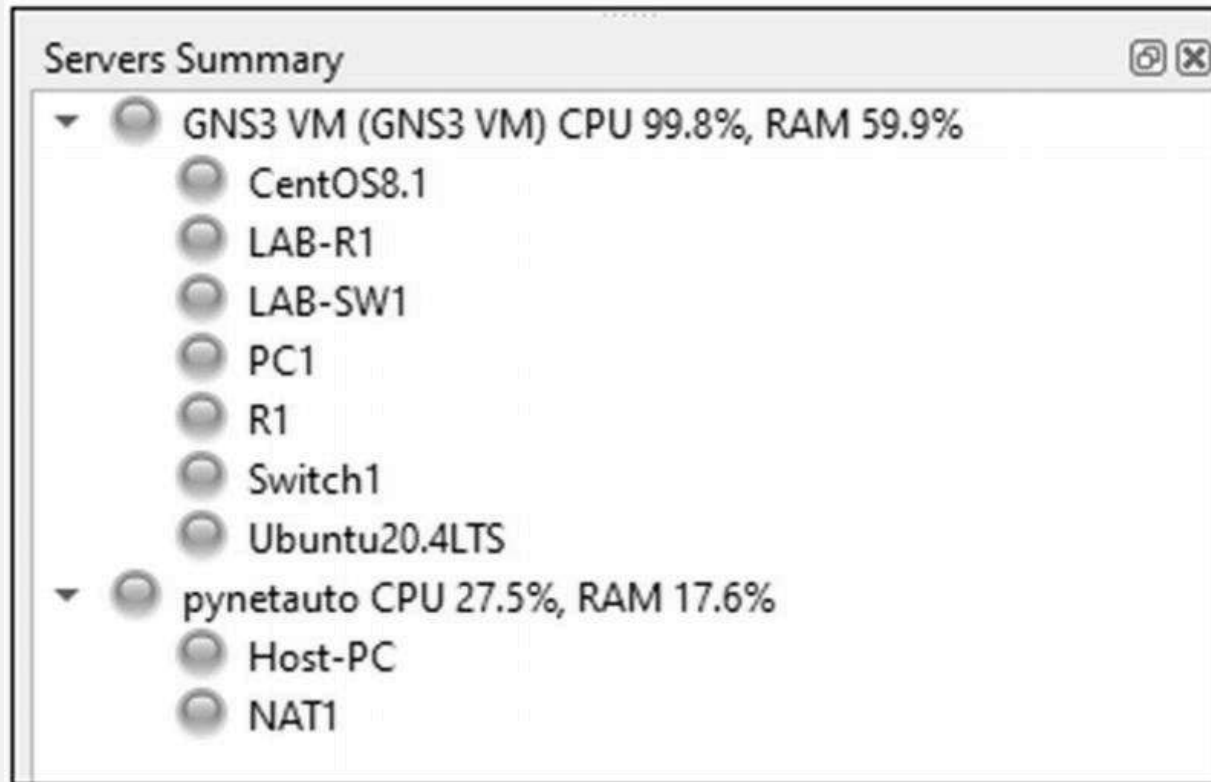


The image shows a 'Topology Summary' window with a table of nodes and their connections. The table has two columns: 'Node' and 'Console'. The nodes are listed with their connections and console access information.

| Node                      | Console                    |
|---------------------------|----------------------------|
| CentOS8.1                 | none                       |
| Host-PC                   | none                       |
| Loopback <=> f0/1 R1      |                            |
| LAB-R1                    | telnet 192.168.65.128:5008 |
| Gi0/0 <=> Gi0/1 LAB-SW1   |                            |
| LAB-SW1                   | telnet 192.168.65.128:5001 |
| Gi0/0 <=> e2 Switch1      |                            |
| Gi0/1 <=> Gi0/0 LAB-R1    |                            |
| NAT1                      | none                       |
| nat0 <=> e0 Switch1       |                            |
| PC1                       | telnet 192.168.65.128:5006 |
| e0 <=> e7 Switch1         |                            |
| R1                        | telnet 192.168.65.128:5005 |
| f0/0 <=> e1 Switch1       |                            |
| f0/1 <=> Loopback Host-PC |                            |
| Switch1                   | none                       |
| e0 <=> nat0 NAT1          |                            |
| e1 <=> f0/0 R1            |                            |
| e2 <=> Gi0/0 LAB-SW1      |                            |
| e7 <=> e0 PC1             |                            |
| Ubuntu20.4LTS             | none                       |

Figure 12-32. Topology Summary window

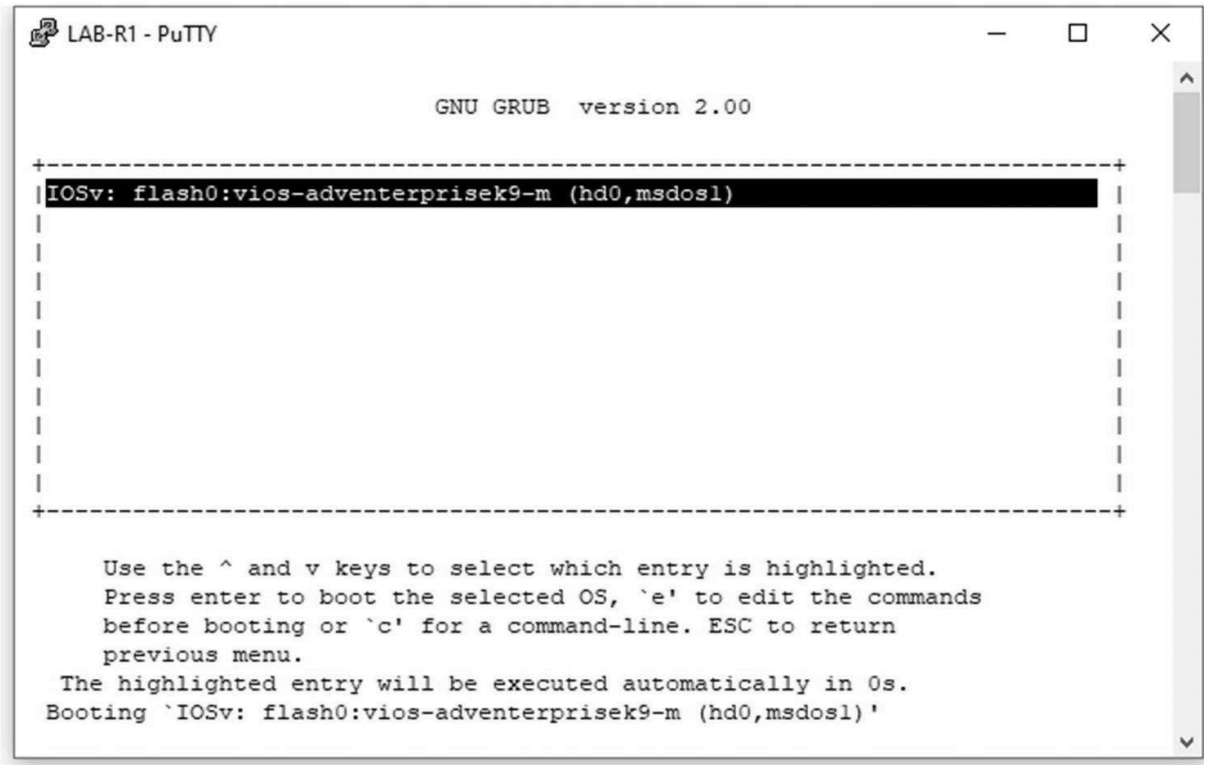
# Task



**Figure 12-33.** Servers Summary window

- 14 When the router boots up, it will display the vIOS version during the bootup. See Figure [12-34](#).

## # Task



**Figure 12-34.** CML L3 router installation, powered-on CML router screen

**15** If you see a console screen similar to Figure [12-35](#), it means you have completed the CML-PERSONAL L3 router on GNS3. CML-PERSONAL is almost identical to Cisco IOS 15.x. Now you are ready for some routing labs using a newer IOS image .

```
Cisco IOS Software, IOSv Software (V IOS-ADVENTERPRISEK9-M), Version
15.6(2)T, RELEASE SOFTWARE (fc2)
```

```
Technical Support: http://www.cisco.com/techsupport
```

```
Copyright (c) 1986-2016 by Cisco Systems, Inc.
```

```
Compiled Tue 22-Mar-16 16:19 by prod_rel_team
```

```
*Jan  9 11:12:35.500: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is OFF
```

```
*Jan  9 11:12:35.501: %CRYPTO-6-GDOI_ON_OFF: GDOI is OFF
```

## # Task

```
*Jan  9 11:12:37.708: %SYS-3-CPUHOG: Task is running for (1999)msecs, mor
than (2000)msecs (0/0),process = Crypto CA.
```

```
*****
```

```
* IOSv is strictly limited to use for evaluation, demonstration and IOS
```

```
* education. IOSv is provided as-is and is not supported by Cisco's
```

```
* Technical Advisory Center. Any use or disclosure, in whole or in part,
```

```
* of the IOSv Software or Documentation to any third party for any
```

```
* purposes is expressly prohibited except as otherwise authorized by
```

```
* Cisco in writing
```

```
.
```

```
*
```

```
*****
```

```
Router#
```

## Quick Communication Test on CML L3 Router Integration on GNS3

Complete the following tasks to allow the newly installed router to communicate to the rest of the devices in your topology:

## # Task

- 1 Quickly log in to the new router and configure the correct router name. Also, configure a valid IP address from the same subnet as VMnet8 (192.168.183.0/24) to interface with GigbitEthernet 0/0. Please note that your subnet is maybe different. Make sure you issue the `no shut` command to bring up the interface.

**# Task**

```
Router#configure terminal # Enter Global Configuration
mode
```

```
Router(config)#hostname LAB-R1 # Rename hostname
```

```
LAB-R1(config)#interface GigabitEthernet 0/0 # Enter
interface Gi0/0 config mode
```

```
LAB-R1(config-if)#ip add 192.168.183.10 255.255.255.0 #
Configure an IP address
```

```
LAB-R1(config-if)#no shut # Bring up this interface
```

```
LAB-R1(config-if)#end # Go back to Privileged EXEC mode
```

On the Lab-R1 router , once the Gi0/0 interface comes up, start sending ICMP packets to the following IP addresses to ensure that LAB-R1 can communicate to all the other devices in our topology.

- 2 VMnet8's default gateway IP (192.168.183.2), PC1's IP address (192.168.183.129), R1's f0/0 (192.168.183.133), the `centos8s1` server (192.168.183.130), and the `ubuntu20s1` server (192.168.183.132).

```
LAB-R1#ping 192.168.183.2 # To Default Gateway
```

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.183.2, time-out is 2 seconds:

```
.!!!!
```

Success rate is 80 percent (4/5), round-trip  
min/avg/max = 8/10/11 ms

## # Task

```
LAB-R1#ping 192.168.183.129 # To PC1
```

Type escape sequence to abort.

```
Sending 5, 100-byte ICMP Echos to 192.168.183.129,  
timeout is 2 seconds:
```

```
.!!!!
```

```
Success rate is 80 percent (4/5), round-trip  
min/avg/max = 7/9/12 ms
```

```
LAB-R1#ping 192.168.183.133 # To R1's f0/0 interface
```

Type escape sequence to abort.

```
Sending 5, 100-byte ICMP Echos to 192.168.183.133,  
timeout is 2 seconds:
```

```
.!!!!
```

```
Success rate is 80 percent (4/5), round-trip  
min/avg/max = 12/15/18 ms
```

```
LAB-R1#ping 192.168.183.130 # To centos8s1 server
```

Type escape sequence to abort.

```
Sending 5, 100-byte ICMP Echos to 192.168.183.130,  
timeout is 2 seconds:
```

```
.!!!!
```

## # Task

```
Success rate is 80 percent (4/5), round-trip  
min/avg/max = 8/8/11 ms
```

```
LAB-R1#ping 192.168.183.132 # To ubuntu20s1 server
```

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 192.168.183.132,  
timeout is 2 seconds:
```

```
.!!!!
```

```
Success rate is 80 percent (4/5), round-trip  
min/avg/max = 8/9/10 ms
```

For verification purposes only, we can also quickly test the SSH  
**3** login to the CentOS8.1 server from LAB-R1 using the `ssh -l`  
`username server_IP` command .

```
LAB-R1#ssh -l pynetauto 192.168.183.130 # SSH login to  
centos8s1 server
```

```
Password:
```

```
Activate the web console with: systemctl enable --now  
cockpit.socket
```

```
Last login: Sat Jan  9 00:00:39 2021 from  
192.168.183.1
```

```
[pynetauto@centos8s1 ~]$ exit
```

```
logout
```

## # Task

```
[Connection to 192.168.183.130 closed by foreign host]
```

```
LAB-R1#
```

Please note that the SSH login to the `ubuntu20s1` server will not work due to a security algorithm mismatch. The SSH login sessions will be initiated from the server side to networking devices, and you do not have to worry about this at this stage.

4

Once you have completed the communication test, save the running configuration of LAB-R1 to the startup configuration .

```
LAB-R1#copy running-config startup-config
```

```
Destination filename [startup-config]?
```

```
Building configuration...
```

```
[OK]
```

## Building a CML-PERSONAL Lab Topology

Before moving onto the actual lab task in the next chapter, let's add a few more devices to the `cmlab-basic` GNS3 project and complete our topology. This step assumes you are continuing the topology build from the last steps.

## # Task

- Continuing from the previous section, add one more IOSvL2 switch and one more IOSv router to the topology as shown in Figure [12-35](#). Rename the switch to `lab-sw2` in lowercase and the router to `lab-r2` also in lowercase. The device names are not capitalized and standardized on purpose here, although it is a best practice to follow a strict naming convention for your network devices. In production, this convention is not followed by every engineer and becomes an annoyance. Hence, to emulate the imperfect production environment, we will not follow a strict device naming convention. Connect Gi0/0 of

# Task

lab-sw2 to Gi0/2 of LAB-SW1 and Gi0/1 of lab-sw2 to Gi0/1 of lab-r2. Also, connect Gi0/0 of lab-r2 to Gi0/1 of LAB-R1.

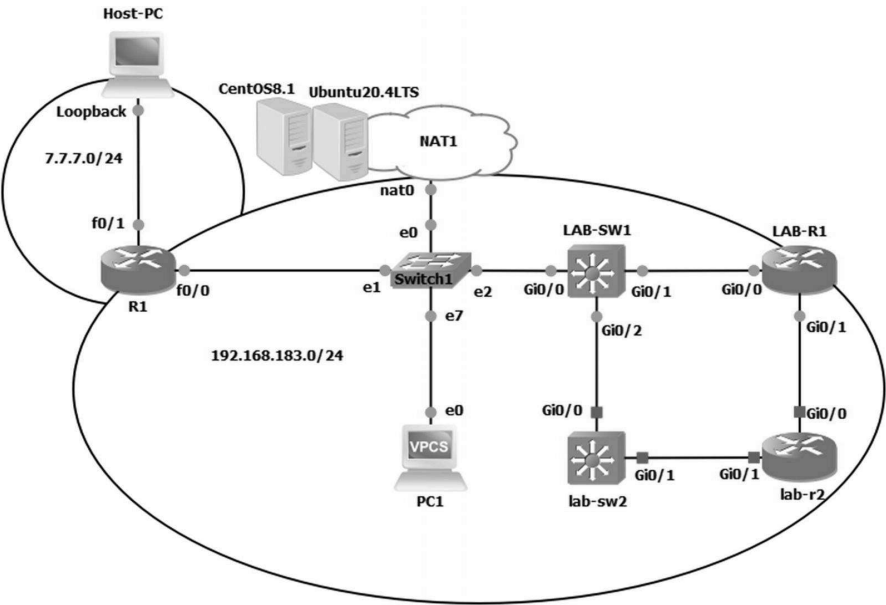


Figure 12-35. Configuring the CML lab topology, one of three

## # Task

Now power on lab-sw2 and lab-r2 as shown in Figure 12-36.

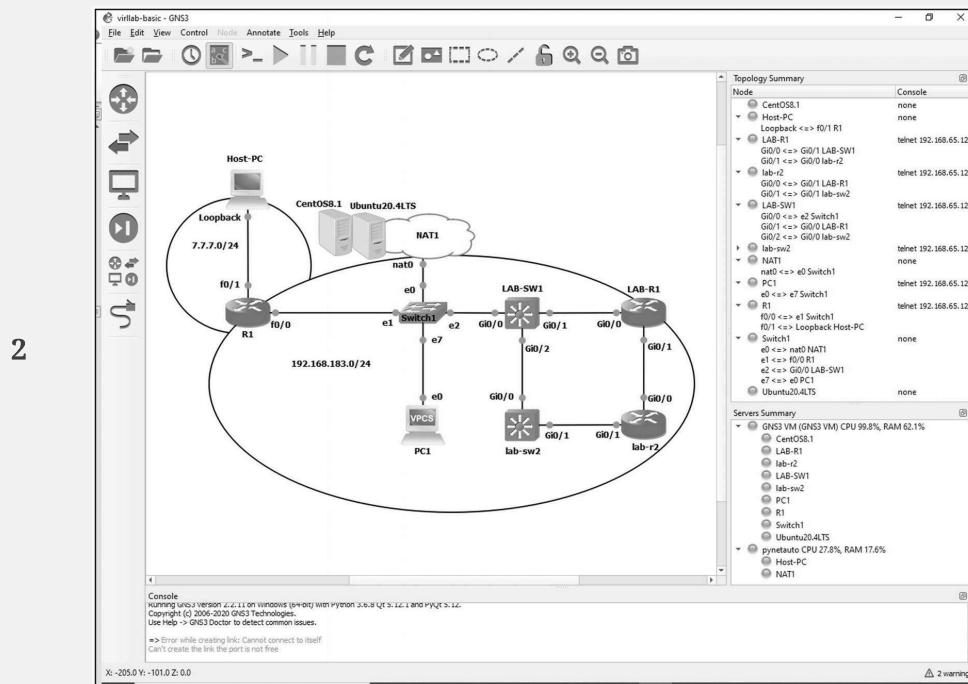


Figure 12-36. Configuring CML lab topology, two of three

At this stage, all devices, including the centos8s1 IP Services server and ubuntu20s1 Python server, should be powered on and running smoothly. Perform ping tests between the GNS3 network and the Linux servers to ensure that all devices can reach others on the VMnet8 network.

4 Here's the R1 IOS router configuration :

First, on R1 (IOS router), let's open the console to assign the IP address manually so its IP address does not change every time the DHCP lease time elapses.

```
R1#show ip interface brief # show interface
configuration
```

## # Task

| Interface | IP-Address | OK? | Method |
|-----------|------------|-----|--------|
| Status    | Protocol   |     |        |

|                 |                 |     |    |
|-----------------|-----------------|-----|----|
| FastEthernet0/0 | 192.168.183.133 | YES |    |
| DHCP            | up              |     | up |

|                 |         |     |    |
|-----------------|---------|-----|----|
| FastEthernet0/1 | 7.7.7.2 | YES |    |
| NVRAM           | up      |     | up |

|                 |                       |     |      |
|-----------------|-----------------------|-----|------|
| FastEthernet1/0 | unassigned            | YES |      |
| NVRAM           | administratively down |     | down |

R1#configure terminal

Enter configuration commands, one per line. End with  
CNTL/Z.

R1(config)#**interface FastEthernet0/0** # Enter interface  
configuration mode

R1(config-if)#**no ip address dhcp** # Disable DHCP IP al-  
location from DHCP server

R1(config-if)#**ip address 192.168.183.133**  
**255.255.255.0** # Manually assign IP Address

R1(config-if)#**speed 100** # Fix speed if you have not  
done so yet above

R1(config-if)#**duplex full** # Fix duplex if you have not  
done so yet above

R1(config-if)#**exit** # Move back to Global configuration  
mode

## # Task

```
R1(config)#ip domain-lookup # Enable DNS lookup
```

```
R1(config)#ip name-server 192.168.183.2 # Assign  
Default Gateway as the name-server
```

```
R1(config)#ip route 0.0.0.0 0.0.0.0 192.168.183.2 #  
Add a default route
```

```
R1(config)#router ospf 1 # Configure basic ospf
```

```
R1(config-router)#network 0.0.0.0 255.255.255.255  
area 0 # add all network to ospf area 0
```

```
R1(config-router)#end # Move back to Exec Privilege  
mode
```

```
R1#write memory # Save current running configuration  
to startup configuration
```

After saving the configuration, perform a quick ping test to the default gateway (192.168.183.2) and any other devices on the network. They should all be reachable from R1.

## 5 Here's the LAB-SW1 configuration :

Now open the LAB-SW1 console and reconfigure the first CML-PERSONAL switch. This switch already has a VLAN 1 interface manually configured, so there's no need to reconfigure the IP address. Since the switches will be used as L2 devices, there is no need to configure DNS or static routes.

```
LAB-SW1#show ip interface brief | be Vlan1
```

## # Task

```
Vlan1          192.168.183.101 YES
NVRAM  up          up
```

Use the following configuration to complete the LAB-SW1 configuration.

```
LAB-SW1#configure terminal # Enter Global configuration mode
```

```
LAB-SW1(config)#enable password cisco123 # Configure enable password
```

```
LAB-SW1(config)#username pynetauto privilege 15 password cisco123 #configure L15 user with pwd
```

```
LAB-SW1(config)#line vty 0 15 # Enter Virtual Terminal Line configuration mode
```

```
LAB-SW1(config-line)#login local # Allow locally configured account
```

```
LAB-SW1(config-line)#transport input all # use both telnet and SSH connection
```

```
LAB-SW1(config-line)#logging synchronous # Change line automatically during log message
```

```
LAB-SW1(config-line)#no exec-timeout # No connection time-out, for Lab use only
```

```
LAB-SW1(config-line)#line console 0 # Enter Line Console 0 config mode
```

## # Task

```
LAB-SW1(config-line)#logging synchronous # Change line
automatically during console message
```

```
LAB-SW1(config-line)#no exec-timeout # No connection
time-out, for Lab use only
```

```
LAB-SW1(config-line)#end # Move back to Exec privilege
mode
```

```
LAB-SW1#copy running-config startup-config # Save
running-config to startup-config
```

```
Destination filename [startup-config]?
```

```
Building configuration...
```

```
[OK]
```

## 6 Here's the lab-sw2 configuration :

Let's apply a similar configuration to the newly added switch, lab-sw2. This switch will also be configured as a layer 2 switch, so there's no need to enable IP routing or to configure DNS. No explanation will be added to this configuration as it is almost identical to the previous LAB-SW2 configuration.

```
Switch>en
```

```
Switch#conf t
```

```
Switch(config)#hostname lab-sw2
```

```
lab-sw2(config)#interface vlan 1
```

## # Task

```
lab-sw2(config-if)#ip add 196.168.183.102  
255.255.255.0
```

```
lab-sw2(config-if)#no shut
```

```
lab-sw2(config-if)#enable password cisco123
```

```
lab-sw2(config)#username pynetauto pri 15 pass  
cisco123
```

```
lab-sw2(config)#line vty 0 15
```

```
lab-sw2(config-line)#login local
```

```
lab-sw2(config-line)#transport input all
```

```
lab-sw2(config-line)#logging synchronous
```

```
lab-sw2(config-line)#no exec-timeout
```

```
lab-sw2(config-line)#line console 0
```

```
lab-sw2(config-line)#logging synchronous
```

```
lab-sw2(config-line)#no exec-timeout
```

```
lab-sw2(config-line)#end
```

```
lab-sw2#copy running-config startup-config
```

```
Destination filename [startup-config]?
```

```
Building configuration...
```

## # Task

[OK]

### 7 Here's the LAB-R1 configuration :

Next, open the console of LAB-R1, assign a DNS server, and then add a static route to allow smooth communication on our lab topology. LAB-R1's Gi0/0 has been manually configured with an IP address of 192.168.183.10/24, so there's no need to configure it. Configure the first CML-PERSONAL router, as shown here:

To configure LAB-R1, use the following configuration:

```
LAB-R1#configure terminal # Enter Global configuration mode
```

```
LAB-R1(config)#interface GigabitEthernet 0/1 # Enter interface configuration mode
```

```
LAB-R1(config-if)#ip address 172.168.1.1 255.255.255.0 # Assign IP address to Gi0/1 interfacing lab-r2
```

```
LAB-R1(config-if)#no shut # Bring up the interface
```

```
LAB-R1(config-if)#exit # Exit interface configuration mode
```

```
LAB-R1(config)#ip domain-lookup # Enable DNS lookup
```

```
LAB-R1(config)#ip name-server 192.168.183.2 # Assign Default Gateway as DNS server
```

```
LAB-R1(config)#ip route 0.0.0.0 0.0.0.0 192.168.183.2 # Add a static Gateway of last resort
```

## # Task

```
LAB-R1(config)#enable password cisco123 # Configure  
enable password
```

```
LAB-R1(config)#username pynetauto privilege 15 pass-  
word cisco123 # configure level 15 user with pwd
```

```
LAB-R1(config)#line vty 0 15 # Enter Virtual Terminal  
Line configuration mode
```

```
LAB-R1(config-line)#login local # Allow locally con-  
figured account
```

```
LAB-R1(config-line)#transport input all # use both  
telnet and SSH connection
```

```
LAB-R1(config-line)#logging synchronous # Change line  
automatically during log message
```

```
LAB-R1(config-line)#no exec-timeout # No connection  
time-out, for Lab use only
```

```
LAB-R1(config-line)#line console 0 # Enter Line  
Console 0 config mode
```

```
LAB-R1(config-line)#logging synchronous # Change line  
automatically during console message
```

```
LAB-R1(config-line)#no exec-timeout # No connection  
time-out, for Lab use only
```

```
LAB-R1(config-line)#end # Move back to Exec privilege  
mode
```

## # Task

```
LAB-R1#copy running-config startup-config # Save running-config to startup-config
```

```
Destination filename [startup-config]?
```

```
Building configuration...
```

```
[OK]
```

## 8 Here's the lab-r2 configuration :

Configure the second CML-PERSONAL L3 router using the following configuration suggestions. The basic configuration is almost identical to that of LAB-R1, so no line-by-line explanation is given.

```
Router#conf t
```

```
Router(config)#hostname lab-r2
```

```
lab-r2(config-if)#interface GigabitEthernet 0/0
```

```
lab-r2(config-if)#ip address 172.168.1.2  
255.255.255.0
```

```
lab-r2(config)#interface GigabitEthernet 0/1
```

```
lab-r2(config-if)#ip add 192.168.183.20 255.255.255.0
```

```
lab-r2(config-if)#no shut
```

```
lab-r2(config-if)#exit
```

```
lab-r2(config)#ip domain-lookup
```

# Task

```
lab-r2(config)#ip name-server 192.168.183.2
```

```
lab-r2(config)#ip route 0.0.0.0 0.0.0.0 192.168.183.2
```

```
lab-r2(config)#enable password cisco123
```

```
lab-r2(config)#username pynetauto pri 15 password  
cisco123
```

```
lab-r2(config)#line vty 0 15
```

```
lab-r2(config-line)#login local
```

```
lab-r2(config-line)#transport input all
```

```
lab-r2(config-line)#logging synchronous
```

```
lab-r2(config-line)#no exec-timeout
```

```
lab-r2(config-line)#line console 0
```

```
lab-r2(config-line)#logging synchronous
```

```
lab-r2(config-line)#no exec-timeout
```

```
lab-r2(config-line)#end
```

```
lab-r2#copy running-config startup-config
```

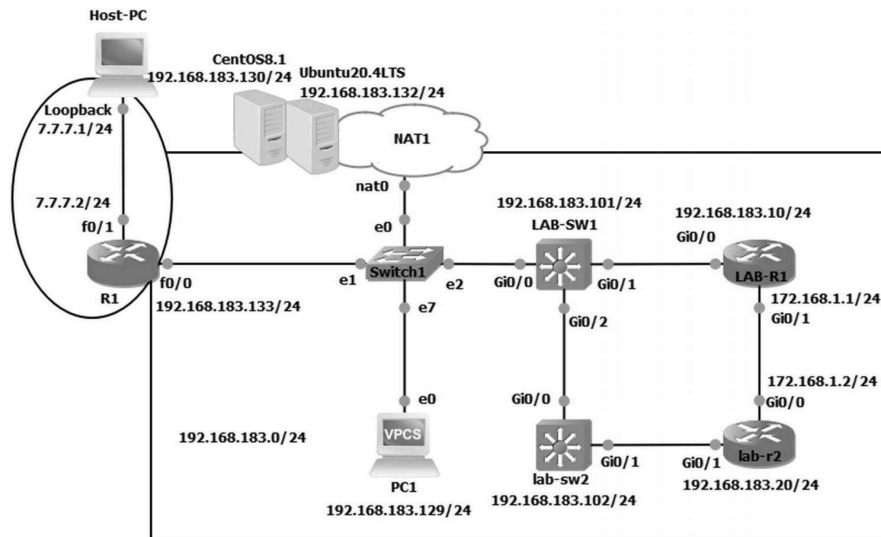
```
Destination filename [startup-config]?
```

```
Building configuration...
```

```
[OK]
```

## # Task

- 9 One of the background shapes has been changed from an oval to a rectangle to make the lab look better, and the topology has been decorated with IP addresses to make the lab steps clear. Take time to decorate your topology to help you with the IP addressing for the coming labs. See Figure [12-37](#), Figure [12-38](#), and Figure [12-39](#).



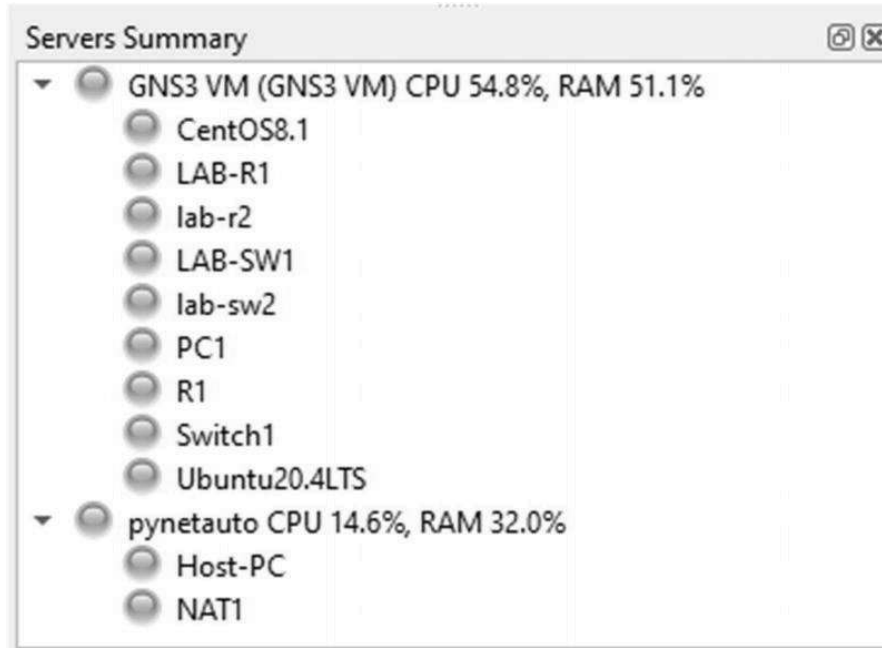
**Figure 12-37.** Configuring the CML lab topology, three out of three

# Task

| Topology Summary          |                         |
|---------------------------|-------------------------|
| Node                      | Console                 |
| CentOS8.1                 | none                    |
| Host-PC                   | none                    |
| Loopback <=> f0/1 R1      |                         |
| LAB-R1                    | telnet 192.168.65.12... |
| Gi0/0 <=> Gi0/1 LAB-SW1   |                         |
| Gi0/1 <=> Gi0/0 lab-r2    |                         |
| lab-r2                    | telnet 192.168.65.12... |
| Gi0/0 <=> Gi0/1 LAB-R1    |                         |
| Gi0/1 <=> Gi0/1 lab-sw2   |                         |
| LAB-SW1                   | telnet 192.168.65.12... |
| Gi0/0 <=> e2 Switch1      |                         |
| Gi0/1 <=> Gi0/0 LAB-R1    |                         |
| Gi0/2 <=> Gi0/0 lab-sw2   |                         |
| lab-sw2                   | telnet 192.168.65.12... |
| NAT1                      | none                    |
| nat0 <=> e0 Switch1       |                         |
| PC1                       | telnet 192.168.65.12... |
| e0 <=> e7 Switch1         |                         |
| R1                        | telnet 192.168.65.12... |
| f0/0 <=> e1 Switch1       |                         |
| f0/1 <=> Loopback Host-PC |                         |
| Switch1                   | none                    |
| e0 <=> nat0 NAT1          |                         |
| e1 <=> f0/0 R1            |                         |
| e2 <=> Gi0/0 LAB-SW1      |                         |
| e7 <=> e0 PC1             |                         |
| Ubuntu20.4LTS             | none                    |

**Figure 12-38.** Topology Summary window

## # Task



**Figure 12-39.** Servers Summary window

10

Now SSH into the `ubutu20s1` or `centos8s1` server and send some ICMP packets to each network device in our topology. Try to send some ICMP messages using the `ping` command . The IP addresses to ping and check connectivity are listed below. Your network IP addresses could be different if you are using a different subnet.

## # Task

```
R1's f0/0      : 192.168.183.133
```

```
PC1           : 192.168.183.129
```

```
LAB-SW1 vlan 1 : 192.168.183.101
```

```
lab-sw2 vlan 1 : 192.168.183.102
```

```
LAB-R1 Gi0/0   : 192.168.183.10
```

```
lab-r2 Gi0/1   : 192.168.183.20
```

```
[pynetauto@centos8s1 ~]$ ping 192.168.183.133 -c 2
```

```
PING 192.168.183.133 (192.168.183.133) 56(84) bytes  
of data.
```

```
64 bytes from 192.168.183.133: icmp_seq=1 ttl=255  
time=7.76 ms
```

```
64 bytes from 192.168.183.133: icmp_seq=2 ttl=255  
time=4.56 ms
```

```
--- 192.168.183.133 ping statistics ---
```

```
2 packets transmitted, 2 received, 0% packet loss,  
time 3ms
```

```
rtt min/avg/max/mdev = 4.556/6.158/7.761/1.604 ms
```

```
[pynetauto@centos8s1 ~]$ ping 192.168.183.129 -c 2
```

## # Task

```
PING 192.168.183.129 (192.168.183.129) 56(84) bytes  
of data.
```

```
64 bytes from 192.168.183.129: icmp_seq=1 ttl=255  
time=31.8 ms
```

```
64 bytes from 192.168.183.129: icmp_seq=2 ttl=255  
time=19.6 ms
```

```
--- 192.168.183.129 ping statistics ---
```

```
2 packets transmitted, 2 received, 0% packet loss,  
time 3ms
```

```
rtt min/avg/max/mdev = 19.562/25.669/31.776/6.107 ms
```

```
[pynetauto@centos8s1 ~]$ ping 192.168.183.101 -c 2
```

```
PING 192.168.183.101 (192.168.183.101) 56(84) bytes  
of data.
```

```
64 bytes from 192.168.183.101: icmp_seq=1 ttl=255  
time=25.0 ms
```

```
64 bytes from 192.168.183.101: icmp_seq=2 ttl=255  
time=24.1 ms
```

```
--- 192.168.183.101 ping statistics ---
```

```
2 packets transmitted, 2 received, 0% packet loss,  
time 2ms
```

```
rtt min/avg/max/mdev = 24.056/24.547/25.038/0.491 ms
```

## # Task

```
[pynetauto@centos8s1 ~]$ ping 192.168.183.102 -c 2
```

```
PING 192.168.183.102 (192.168.183.102) 56(84) bytes  
of data.
```

```
64 bytes from 192.168.183.102: icmp_seq=1 ttl=255  
time=35.5 ms
```

```
64 bytes from 192.168.183.102: icmp_seq=2 ttl=255  
time=33.0 ms
```

```
64 bytes from 192.168.183.20: icmp_seq=1 ttl=255  
time=184 ms
```

```
64 bytes from 192.168.183.20: icmp_seq=2 ttl=255  
time=46.10 ms
```

```
--- 192.168.183.102 ping statistics ---
```

```
2 packets transmitted, 2 received, 0% packet loss,  
time 3ms
```

```
rtt min/avg/max/mdev = 33.049/34.295/35.542/1.260 ms
```

```
[pynetauto@centos8s1 ~]$ ping 192.168.183.10 -c 2
```

```
PING 192.168.183.10 (192.168.183.10) 56(84) bytes of  
data.
```

```
64 bytes from 192.168.183.10: icmp_seq=1 ttl=255  
time=37.8 ms
```

```
64 bytes from 192.168.183.10: icmp_seq=2 ttl=255  
time=20.1 ms
```

## # Task

```
--- 192.168.183.10 ping statistics ---
```

```
2 packets transmitted, 2 received, 0% packet loss,  
time 3ms
```

```
rtt min/avg/max/mdev = 20.081/28.936/37.792/8.857 ms
```

```
[pynetauto@centos8s1 ~]$ ping 192.168.183.20 -c 2
```

```
PING 192.168.183.20 (192.168.183.20) 56(84) bytes of  
data.
```

```
64 bytes from 192.168.183.20: icmp_seq=1 ttl=255  
time=184 ms
```

```
64 bytes from 192.168.183.20: icmp_seq=2 ttl=255  
time=46.10 ms
```

```
--- 192.168.183.20 ping statistics ---
```

```
2 packets transmitted, 2 received, 0% packet loss,  
time 3ms
```

```
rtt min/avg/max/mdev = 46.992/115.296/183.601/68.305  
ms
```

Now you have completed the integration of the CML L3 router image and communication test between the Python server and other networking devices in your topology.

## Summary

In this chapter, you were introduced to Cisco CML integration with GNS3. You downloaded and integrated the CML L2 and L3 images with GNS3; then you built a basic CML lab topology and took it for a quick test-drive.

Then you confirmed the network connectivity between the Python automation server and all the routers and switches in a `cml1ab-basic` topology. You have completed the CML lab preparation and now are ready to jump right in and start writing some Python code to Telnet or SSH and play with the Cisco virtual routers and switches on GNS3.

In Chapter **13**, you will learn to control the network devices from the Python Linux server starting with the Python Telnet library.

Table of contents collapsed